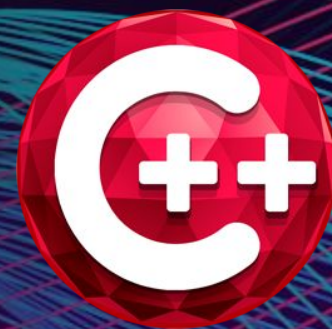




See What's New In
<RAD STUDIO 13 FLORENCE>



Agenda

Delphi: Ternary operator and More

Clang 20 upgrade for C++ Builder

Complete 64-bit IDE for Windows 64-bit platform

VA & code tooling improvements

SmartCore AI Component Pack

VCL TitleBar Styling and new FMX controls

WebBroker and WebStencils extensions





What is RAD Studio?

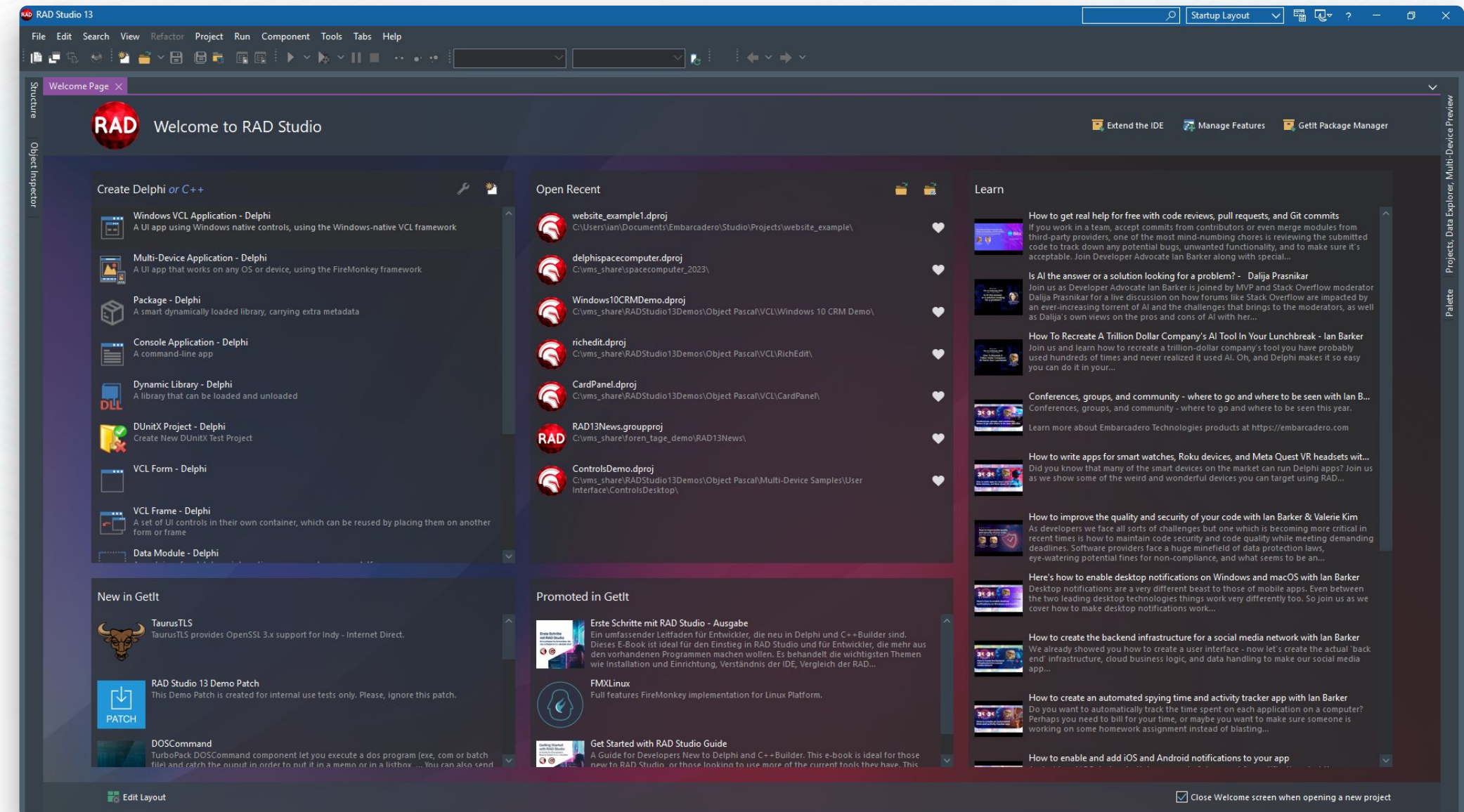
RAD Studio is a software development IDE for building

- Delphi native applications for Windows, macOS, iOS, Android, and Linux from one codebase

- C++ Windows apps

featuring

- visual designers
- embedded AI
- integrated toolchains
- comprehensive data access
- a component-based architecture
- an accelerated development workflow





Programming Languages and Compilers

Delphi Compilers

*Delphi's Object Pascal is a **powerful** and **readable** OOP language*
*Delphi compilers generate **native code** for **9 different targets***

RAD 13 includes significant extensions to the language, Embarcadero is committed on keeping Delphi's Object Pascal modern and relevant



New Delphi Language Features in 13

- Delphi **ternary operator**

- Implemented using the **if** keyword
 - Same of the ternary conditional operator in other programming languages
- A condition in an expression
- Specific rules in terms of the coherence among types of the two sub-expressions
- Notice how other operations have higher priorities: use extra parentheses as needed
- Not only assignments ->

```
procedure TForm31.Button1Click(Sender: TObject);
begin
    var X: Integer := 22;

    if Left < 100 then
        X := 22
    else
        X := 45;
    X := if Left < 100 then 22 else 45;

    ShowMessage (X.ToString);

    ShowMessage (if Left < 100 then 'Small' else 'Big')
end;
```

```
80 procedure TForm31.Button3Click(Sender: TObject);
    begin
        ShowMessage (if Left < 100 then 'Small' else 100)
    end.
85
```

! E2010 Incompatible types: 'string' and 'Integer'



More New Delphi Language Features in 13

- **NameOf** intrinsic function
 - Get a string representation of identifiers in code
- New **{ \$PUSHOPT }** and **{ \$POPOPT }** compiler directives
 - Save and restore the current set of options and warning configurations
 - Similar to C++ “#pragma option push” and “#pragma option pop”
- Implicit "Self" parameter declaration in record class operators Initialize and Finalize (saves typing!)

```
type
  TRec = record
    FIntData: Integer;
    class operator Initialize; // equiv. Initialize(out Self: TRec);
    class operator Finalize; // equiv. Finalize(var Self: TRec);
    procedure Test;
  end;
```




Even More New Delphi Language Features in 13

- **is not** and **not in** operators

```
if not (Obj is TFoo) then ... // classic syntax
if Obj is not TFoo then ... // new syntax
if not (x in [Apple, Orange]) then ... // classic syntax
if x not in [Apple, Orange] then ... // new syntax
```

- **noreturn** directive for procedure declaration
 - Skips W1035 warning: *Return value of function might be undefined*
- Additional constraints for generic types
 - interface
 - unmanaged
 - nonnull
- Extended the **\$IFOPT** directive to support long directives
 - Example: {\$IFOPT SCOPEDENUMS ON}

C++Builder Compilers

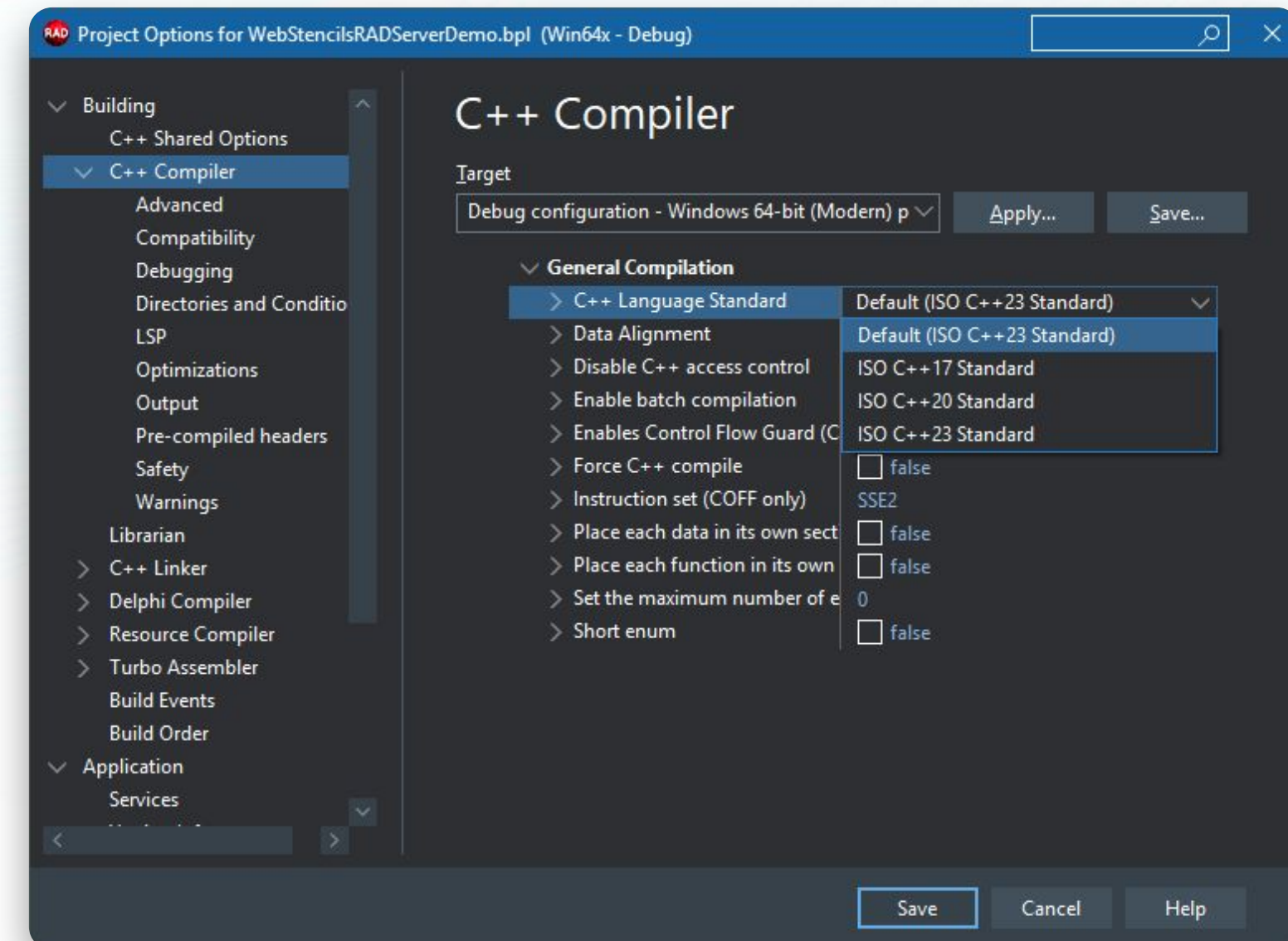
*C++Builder includes enhanced **Clang-based compilers**
The language is extended for **integration** with **Delphi** libraries*

*RAD 13 a new “Win64 modern” compiler **updated to Clang20***



Modern C++ Win64 Compiler to Clang 20

- C++Builder Modern Win64 compiler updated to **Clang 20** (and LLVM 20)
- Clang 20 **introduces C++ 23** support
 - Modern toolchain defaults to C++ 23 (users can choose to target C++20 or C++17 as needed)
 - Details of current clang and libc++ support for specific C++ 23 features at:
 - https://clang.llvm.org/cxx_status.html
 - <https://libcxx.llvm.org/Status/Cxx23.html>
- A significant further upgrade to the C++ Modern toolchain
 - A clear indication of Embarcadero's commitment to C++Builder





Modern C++ Win64 Compiler to Clang 20

Along with the update to LLVM and Clang 20

- Improved Address Sanitizer (ASAN) runtime (Windows 11)
- Versioned shared libraries
 - Shared runtime libraries (e.g., libcplusplus.dll) suffixed with the BDS version number (e.g., libcplusplus-370.dll)
 - Prevents runtime conflicts when multiple versions of RAD Studio are installed
- Improved compatibility with legacy versions of C++ compilers



Installation and IDE

Installation

*RAD Studio has a **two phase installation**:
initial installer and platforms + features configuration
Online installation or **Offline** (ISO)*

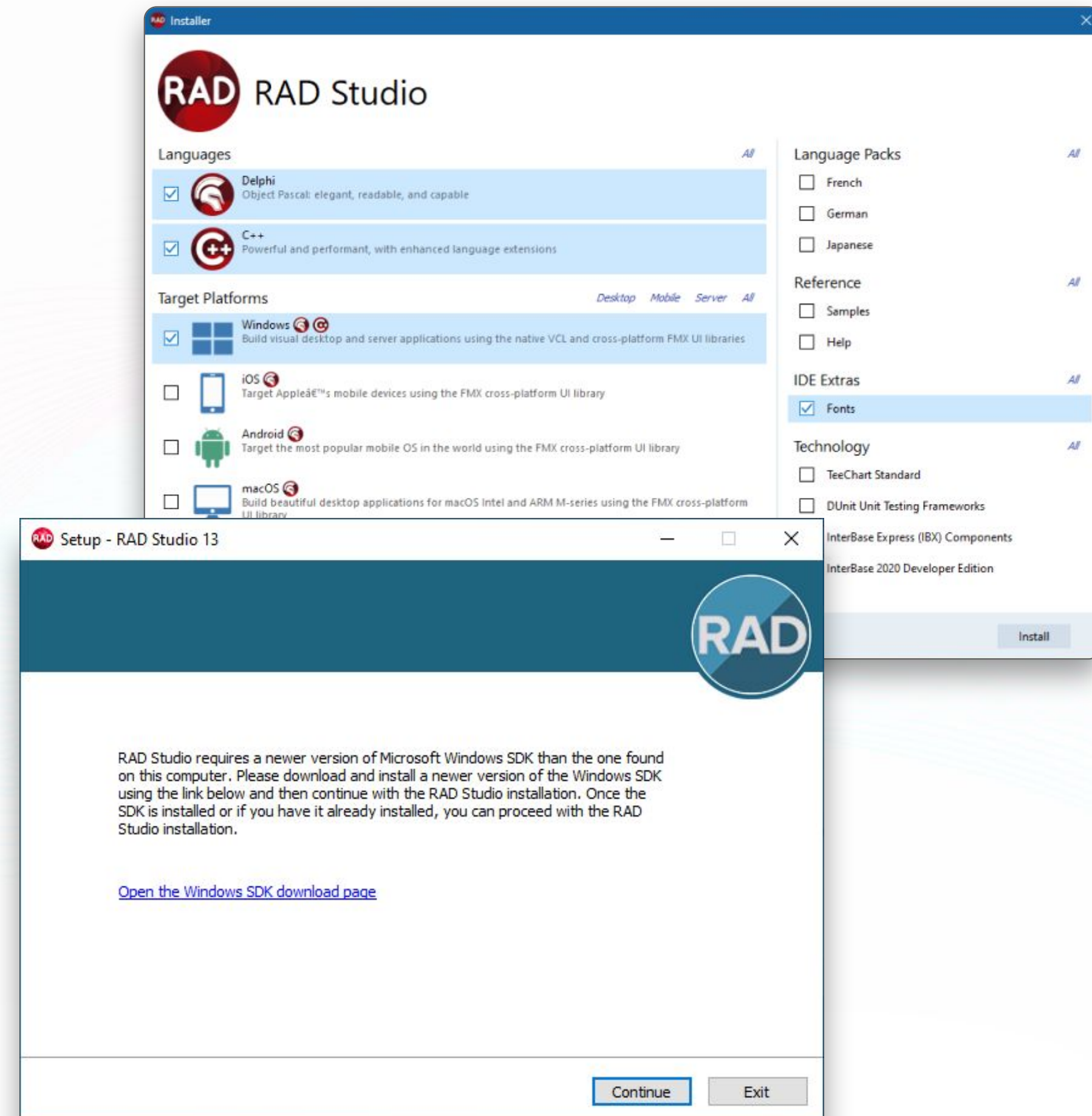
*RAD 13 installer has the **64-bit IDE** as a core feature*



RAD Studio 13 Installation

Platform Installer

- Removed 64-bit IDE as an option (it's now integrated)
- Windows SDK installation is now a prerequisite
 - Install the Windows SDK before installing RAD Studio (you'll get the notice here, if not found)
 - RAD Studio installer alerts if not found, offers download link





Side Note: Unified Internal Versions to 37

- Compiler version is **370**
 - Incremented as usual, all others are now aligned to this
- The main registry entry:
 - HKEY_CURRENT_USER\SOFTWARE\Embarcadero\BDS\37.0
 - The package version and RTL version
 - The FMX version: **FMX.Types.FireMonkeyVersion**
 - The PA Server version
 - The folder where the product is installed by default:
C:\Program Files (x86)\Embarcadero\Studio\37.0



RAD Studio IDE(s)

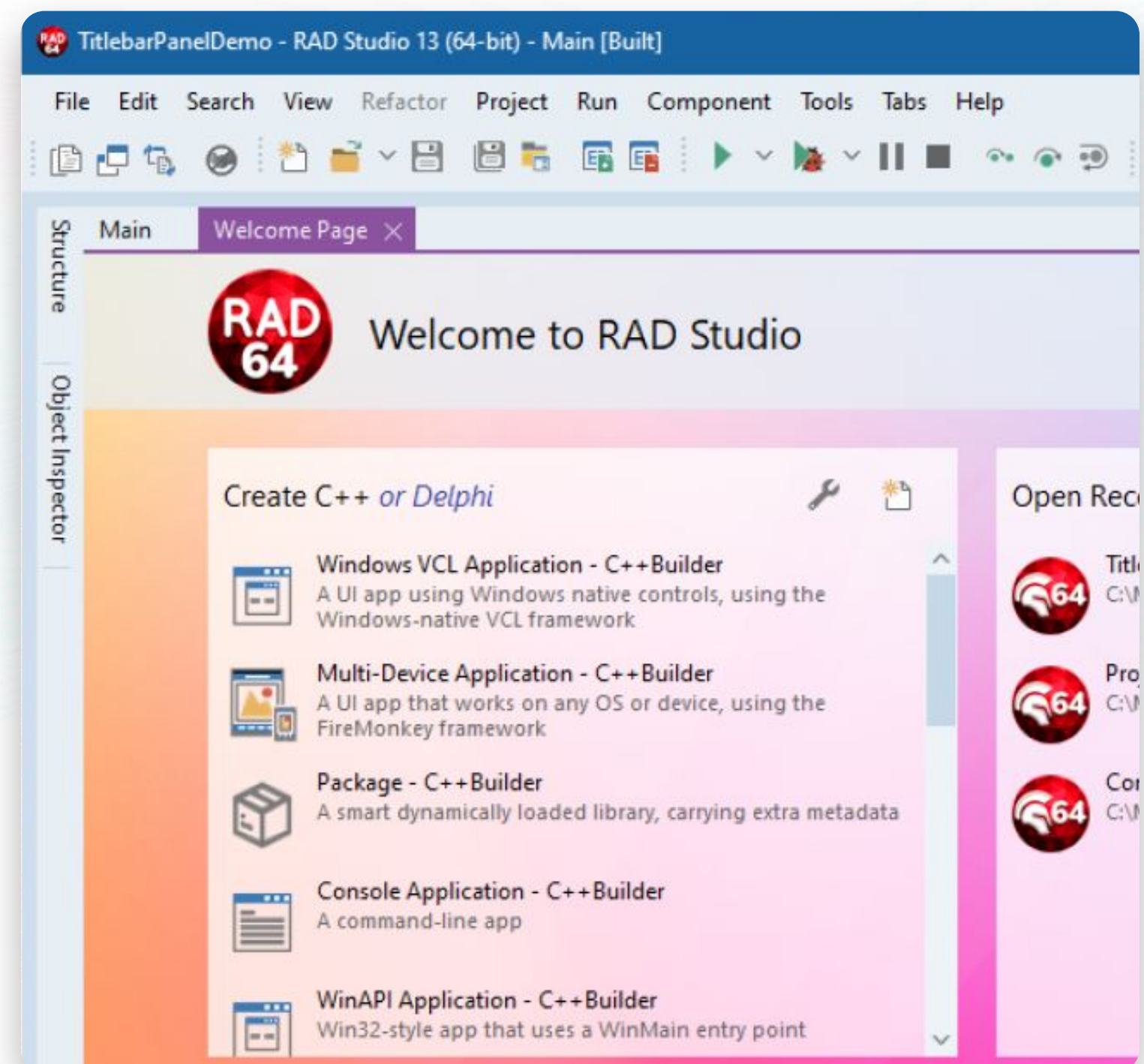
*RAD Studio has a **comprehensive and powerful IDE**
It's also **highly customizable** by customers and third party vendors*

*RAD 13 graduates the **64-bit IDE** for Win64 targets, with the two IDEs
remaining available **in parallel***



Complete 64-bit IDE for Win64

- Complete solution for Win64 target platform
- Now including:
 - Code tooling for C++ via Visual Assist
 - Support for COM, ActiveX and Type libraries
 - Live Templates (Ctrl+J) and the Surround menu
 - Quality improvements for IDE and debugger
- **Removed UML Modeling** (also from 32-bit IDE)
 - And also related features like Audits and Metrics
- **Graduating to a full feature**
 - No separate install option (it's always installed)
 - Not calling it "Initial release" any more

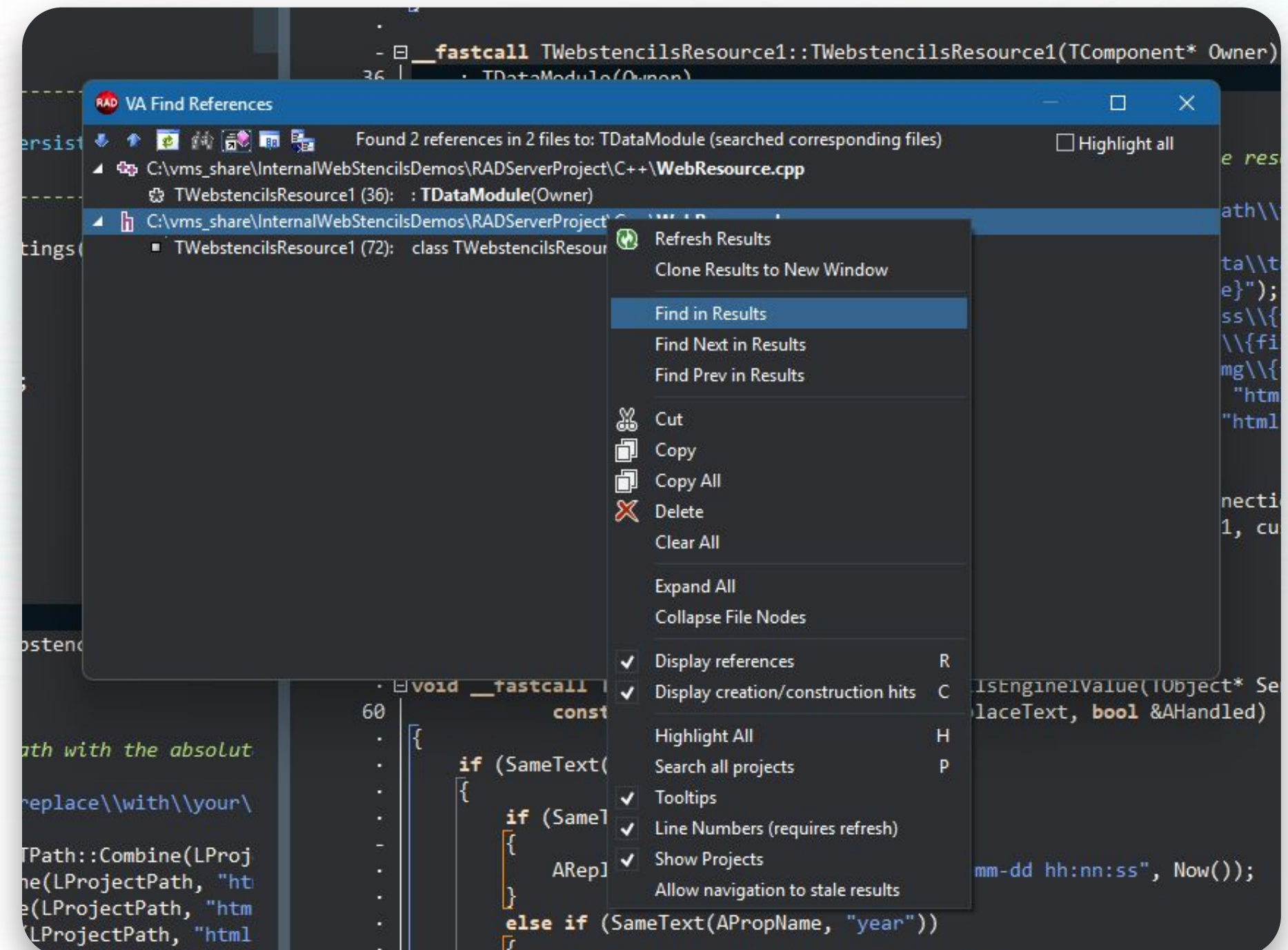




Code Tooling (for C++)

Visual Assist a.k.a. **VA** for C++Builder: Reworked Integration

- Enable VA in the 64-bit IDE
- Improved stability in 32-bit IDE
- Now working when switching projects, also in projects groups with different toolchains and configurations





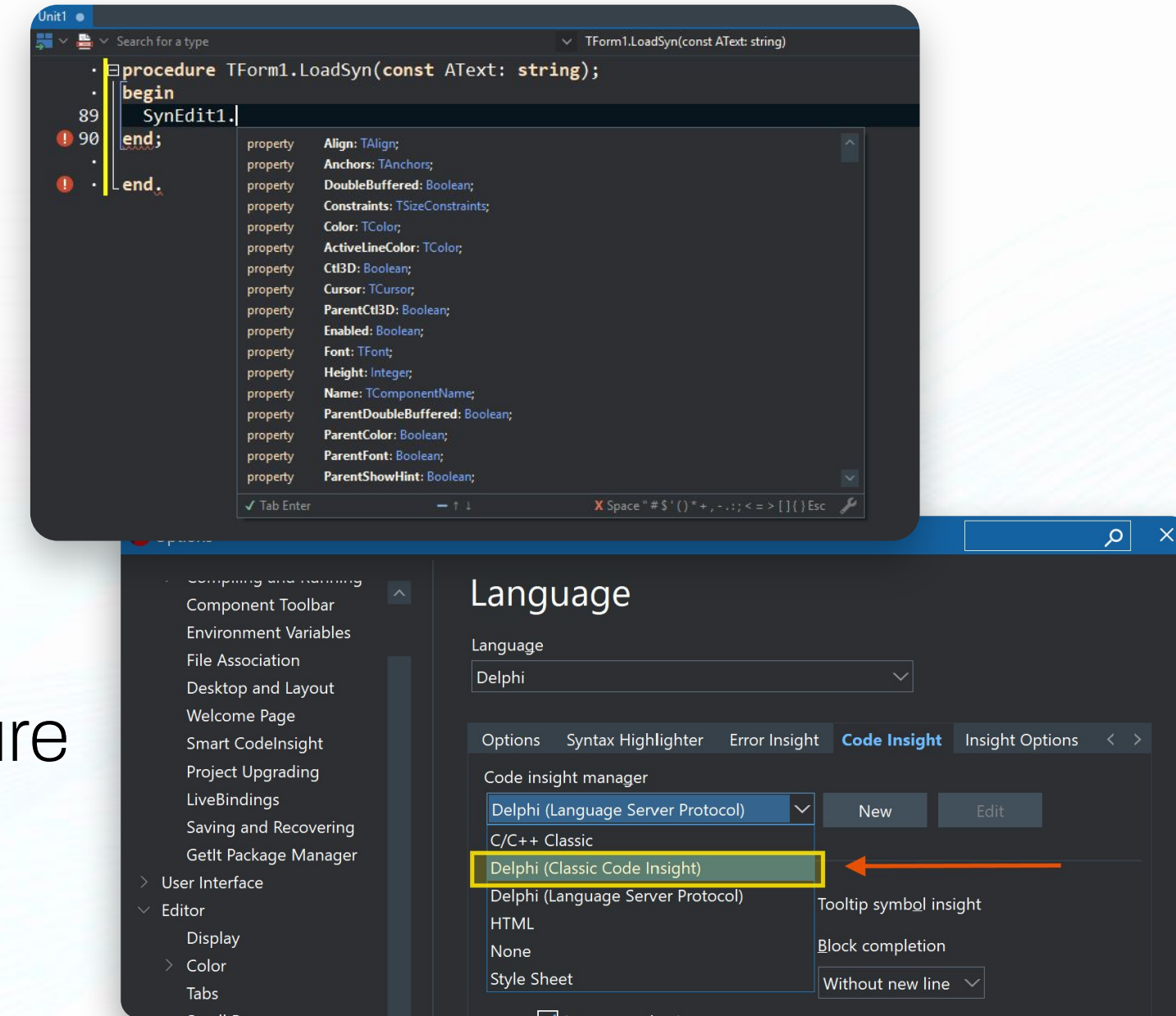
Code Tooling (for Delphi)

- **Delphi Code Insight**

- Old “Classic” Delphi CodeInsight engine for code completion available again
 - Alternative to the DelphiLSP powered version in the IDE
- Continuing the work to improve DelphiLSP architecture

- **New Delphi Upgrade Advisor wizard**

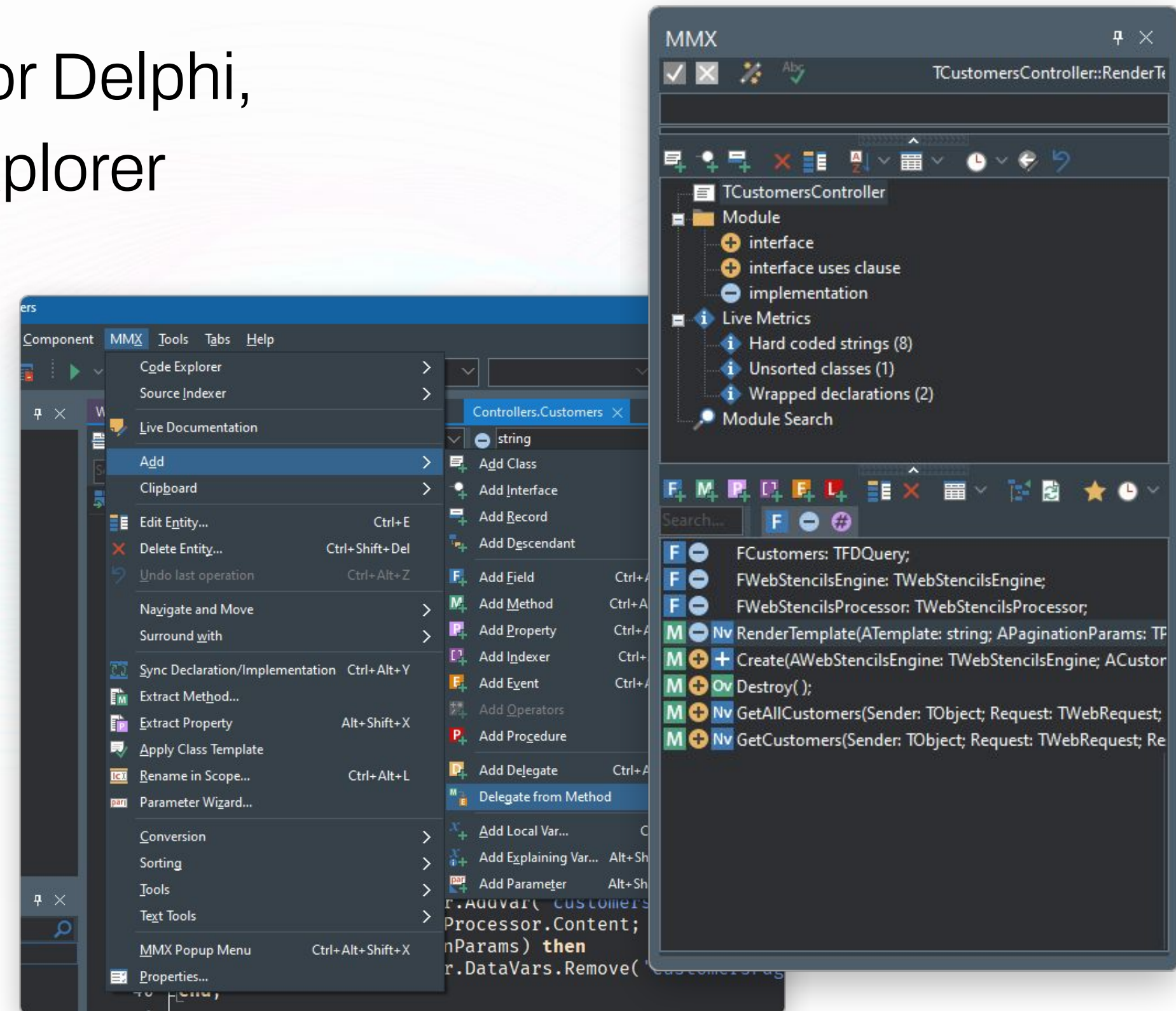
- Allows developers to identify meaningful changes to project configuration and code that can significantly improve the compilation time and code tooling experience.
- In GetIt. Includes scripts for uses statement cleanup





Recommending MMX Code Explorer for Delphi

- While working on new native tooling for Delphi, we promote the use of MMX Code Explorer
 - By MVP Uwe Raabe, available in GetIt
 - See also <https://www.mmx-delphi.de/>
 - For 32-bit and 64-bit IDE
- Offers code navigation, extensive refactoring, a basic formatter and a number of coding helpers, and much, much more





Debugger Quality

- **Moved our LLDB-based debuggers to version 20 of LLDB**
 - Matching the update of the C++ compiler to Clang 20
- **Improved the quality of the debugger**
 - Focusing primarily on the 64-bit IDE debug experience
 - Improved debugger performance
 - In the IDE options, you can enable Code Highlight in the Debug Windows.
 - (C++) argc and argv now correctly handle arguments with spaces when running under the debugger.
 - (C++) Improved how OutputDebugStringW handles Unicode output.
 - (C++) Wide character values that are non-printing are now displayed using the correct format in debugger output.

(List continues on the next slide...)



Debugger Quality

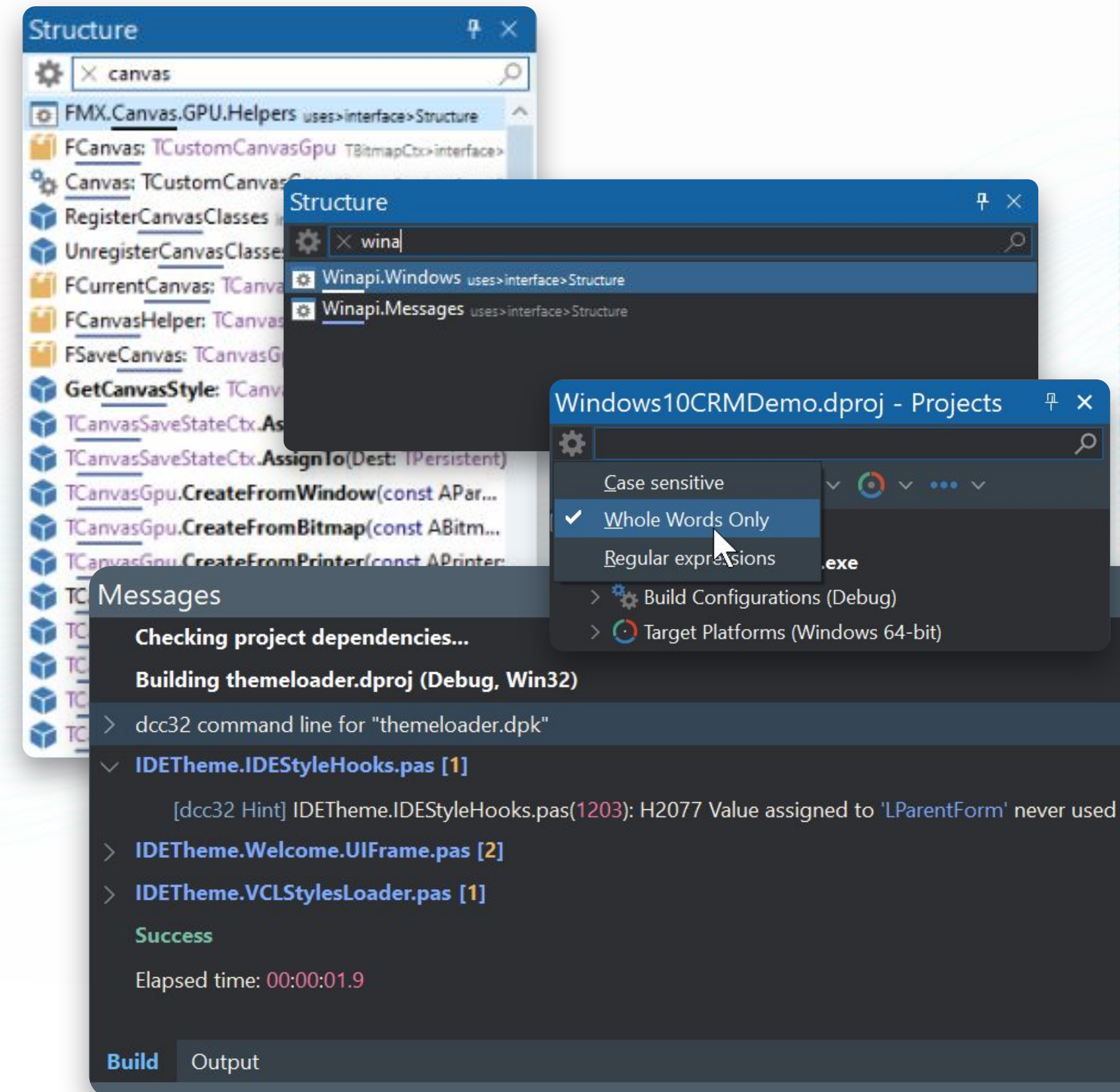
(list continued from previous slide)

- Made pointer addresses look more Delphi-style.
- (Delphi) Scoped enumeration values now remain contained adequately within their declared type, avoiding conflicts in the global namespace.
- (Delphi) Object member set values are now displayed correctly during debugging.
- (Delphi) Scoped enumeration values are now rendered accurately in debugger views.
- (Delphi) Class variable names are now correctly resolved, with improved name mangling handling.
- (Delphi) Unicode variable handling has been improved; the debugger now accepts a broader range of Unicode identifiers.
- **Attach to Process dialog box redesigned to include additional columns that display information about processes**



New Search Features in IDE

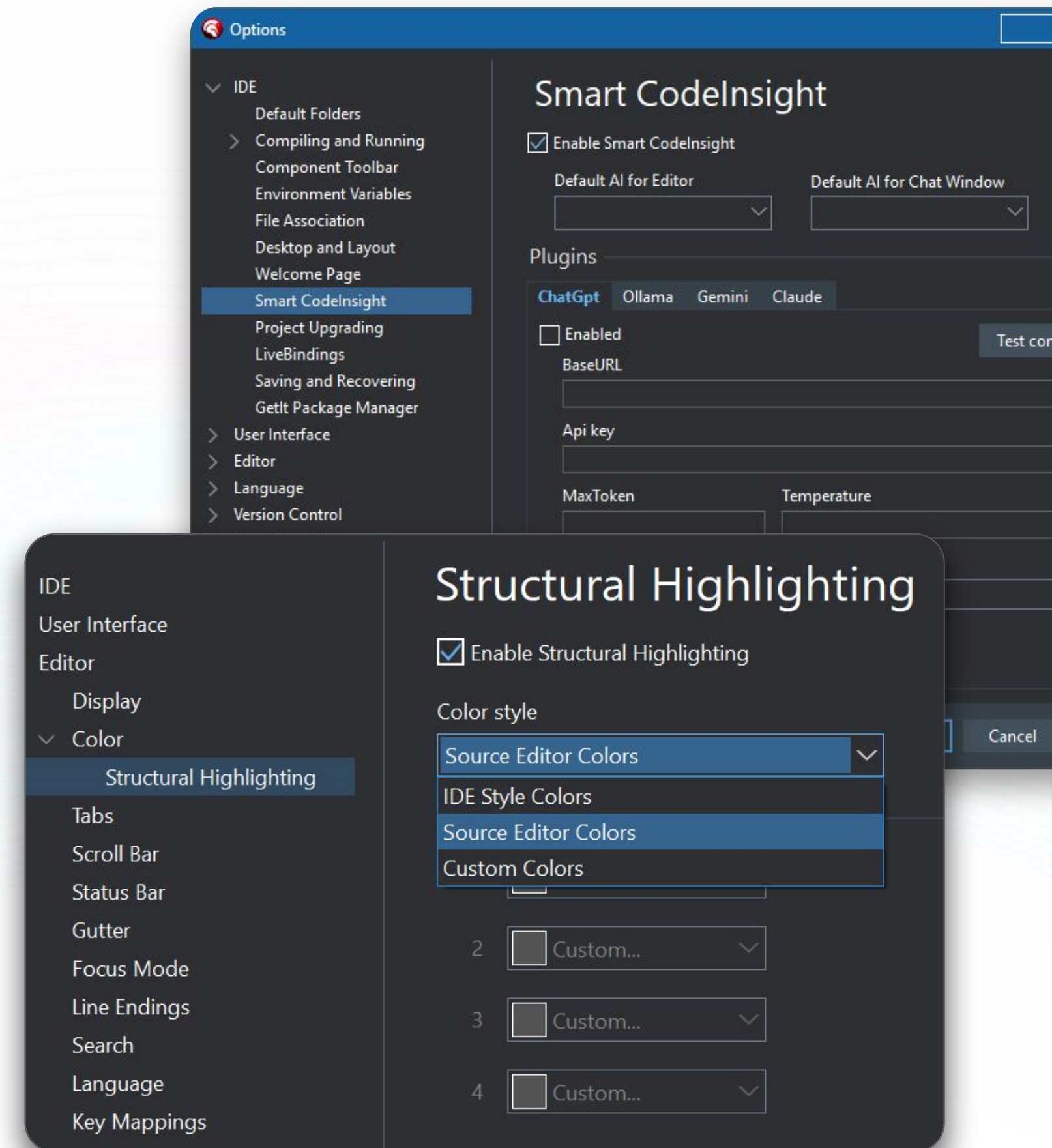
- Extensive and configurable search features in many IDE panes
 - Project Manager
 - Structure View
 - Message pane and Events log
- For the Message Log
 - Highlighting options
 - Grouping Hints and Warning by source code file





IDE Quality

- Improvements to Smart CodeInsight (our AI for developers)
 - New Test Connection feature in the configuration dialog
 - Improvements to the prompts and the response processing
- Option to use the editor background color for editor tabs
- Activate the IDE theme for Structural Highlight colors
- Copy error messages from the structure pane
- Stack trace visible by default in the IDE exception dialog





IDE: ToolsAPI

- *RAD Studio ToolsAPIs allow customers and third party vendors to expand and customize the RAD Studio IDE via official interfaces*
- ToolsAPI extensions in 13.0 include:
 - Welcome Page method to refresh the UI
 - Notifications when the Editor Cursor Moves or the User Types
 - Cursor Moved Event
 - Key Pressed Events

ToolsAPI

[Up to Parent: Unit List](#)

Units [\[edit\]](#)

[D:\C ToolsAPI](#)

[D:\C ToolsAPI.AI](#)

[D:\C ToolsAPI.Editor](#)

[D:\C ToolsAPI.WelcomePage](#)

Classes [\[edit\]](#)

[D:\C ENonAIRException](#)

[D:\C EPersonalityException](#)

[D:\C IBorlandIDEServices](#)

[D:\C IBorlandIDEServices70](#)

[D:\C INTAAddInOptions](#)

[D:\C INTAComponent](#)

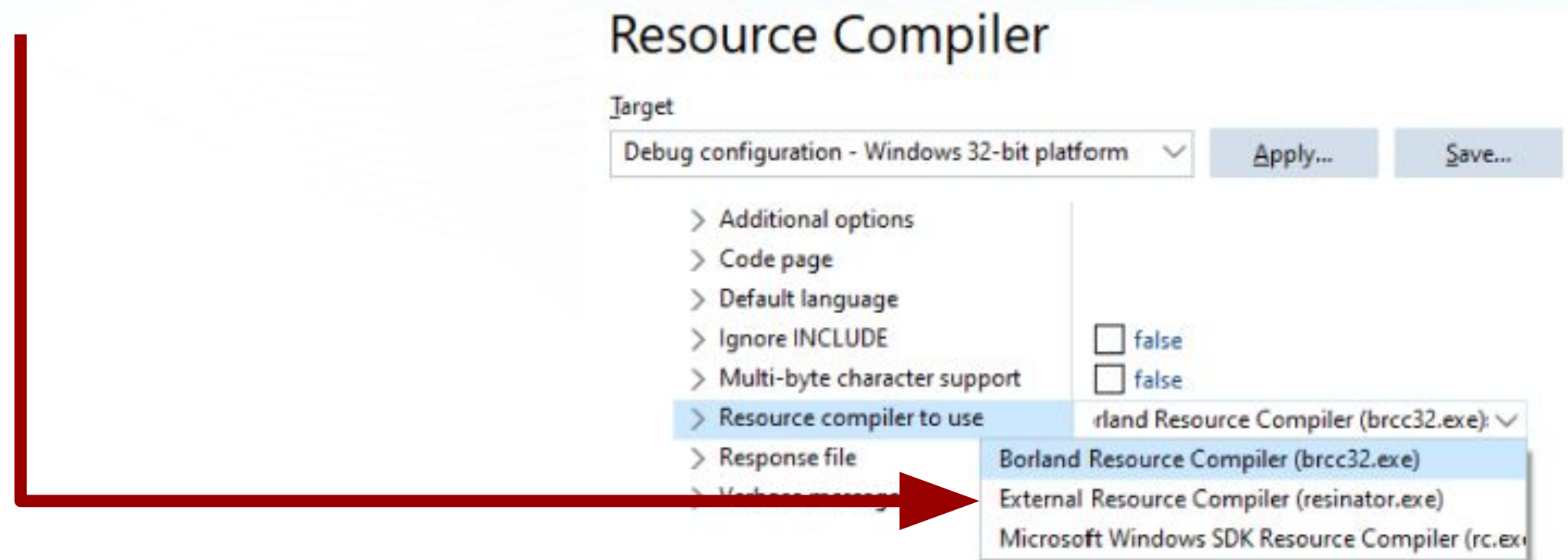
[D:\C INTACustomDockableForm](#)

[D:\C INTACustomDrawCodeInsightViewer](#)



Resource Compiler

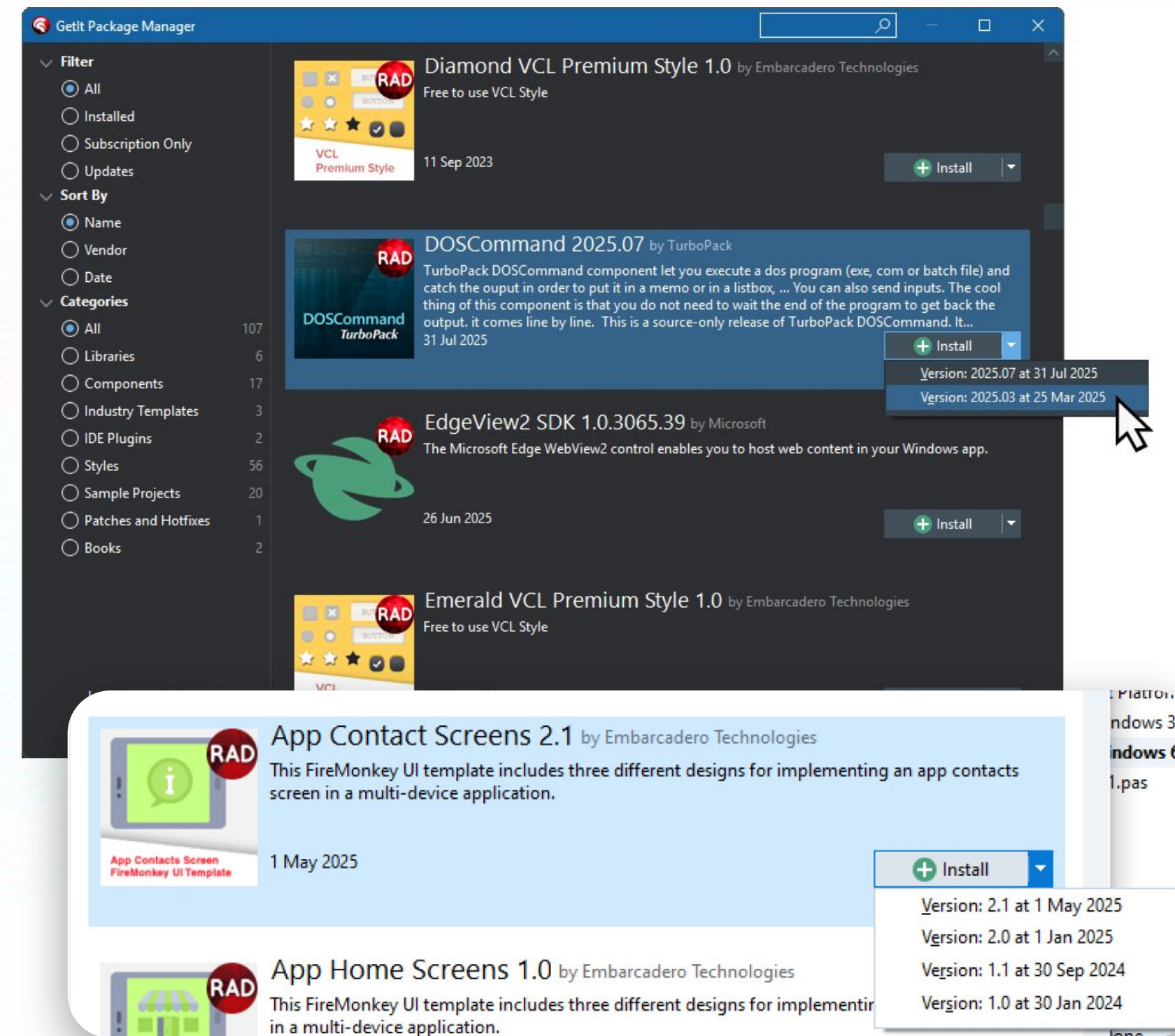
- RAD 13 includes an additional, open-source resource compiler called **resinator** (see <https://github.com/squeek502/resinator>)
 - This is the default in the build system
 - We are making it easier for developers to pick a different resource compiler, including the classic resource compiler that comes with the Windows SDK





GetIt Packages Versioning

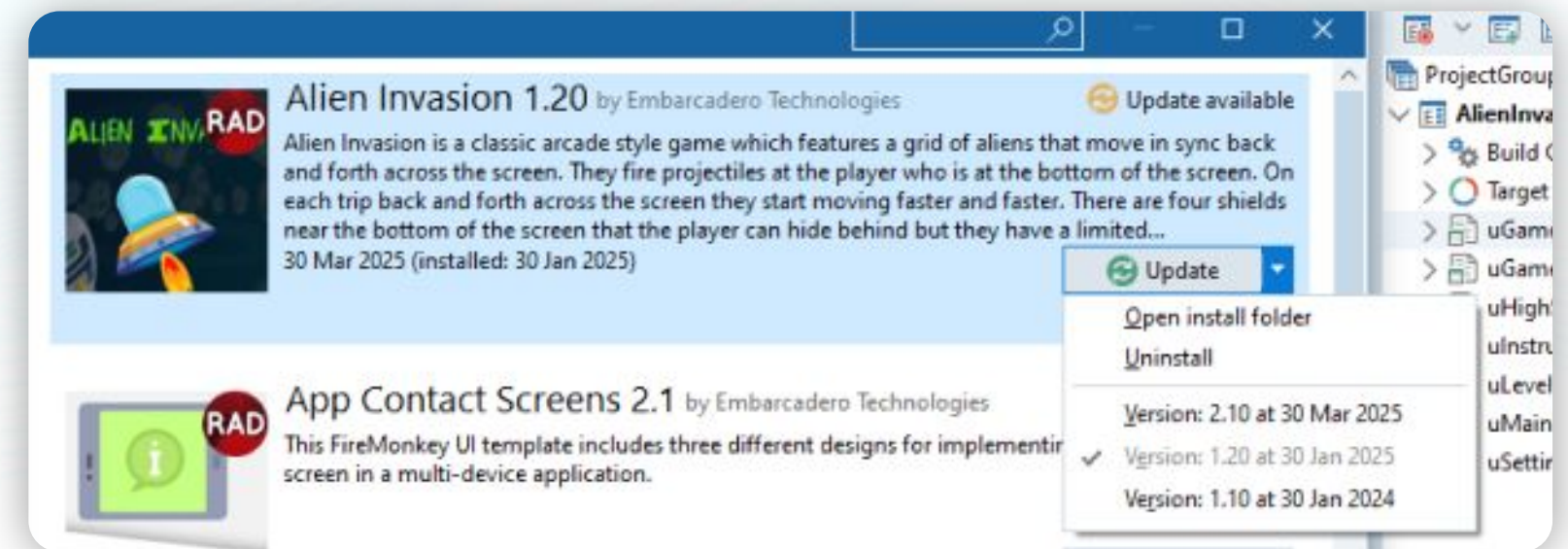
- GetIt is RAD Studio's package manager
 - Offers additional features and addons by Embarcadero
 - Includes open source and other libraries
- In 13, GetIt adds versioning support
 - Offers multiple versions of the same package
 - Users can pick a specific version
 - Rather than be forced to install the most recent version
 - Project dependencies are also package version specific





Other Changes in GetIt UI

- GetIt UI now features a general-purpose command button for the main operation, along with a drop-down menu offering additional features.
- New command to open the package installation folder in Windows Resource Explorer
- Ability to export the Installed Packages (to a string, to be used GetItCmd)





New “RAD AI Companion” Website

- Documentation is available as usual:
 - Integrated help
 - Online docwiki (<https://docwiki.embarcadero.com/RADStudio/>)
- Embarcadero is now launching a **RAD Studio tailored chat bot**, offering the ability to
 - Query an *AI specifically* trained on our product and its libraries
 - Ask it to generate application source code
 - Available at <https://www.embarcadero.com/radaicompanion>



RAD AI Companion

See next slide for the actual UI

Home > Resources > RAD AI Companion

I'm an AI-powered chatbot here to help you get the most out of Delphi, C++Builder and RAD Studio. I can answer documentation questions, generate code examples, and suggest possible fixes.

By using the RAD AI Companion, please note that your questions and responses may be recorded and shared with third parties for analysis and improvement.

Information provided is for guidance only and should be independently reviewed and verified.

What's the most important class in the VCL library?



The most important class in the VCL library is `TObject` . All classes in the VCL descend from `TObject` , which introduces methods that implement fundamental behavior like construction, destruction, and message handling. When you write classes of your own, they should descend from `TObject` in the class library you plan to use.

For more information, you can visit the official documentation: [System.TObject](#).

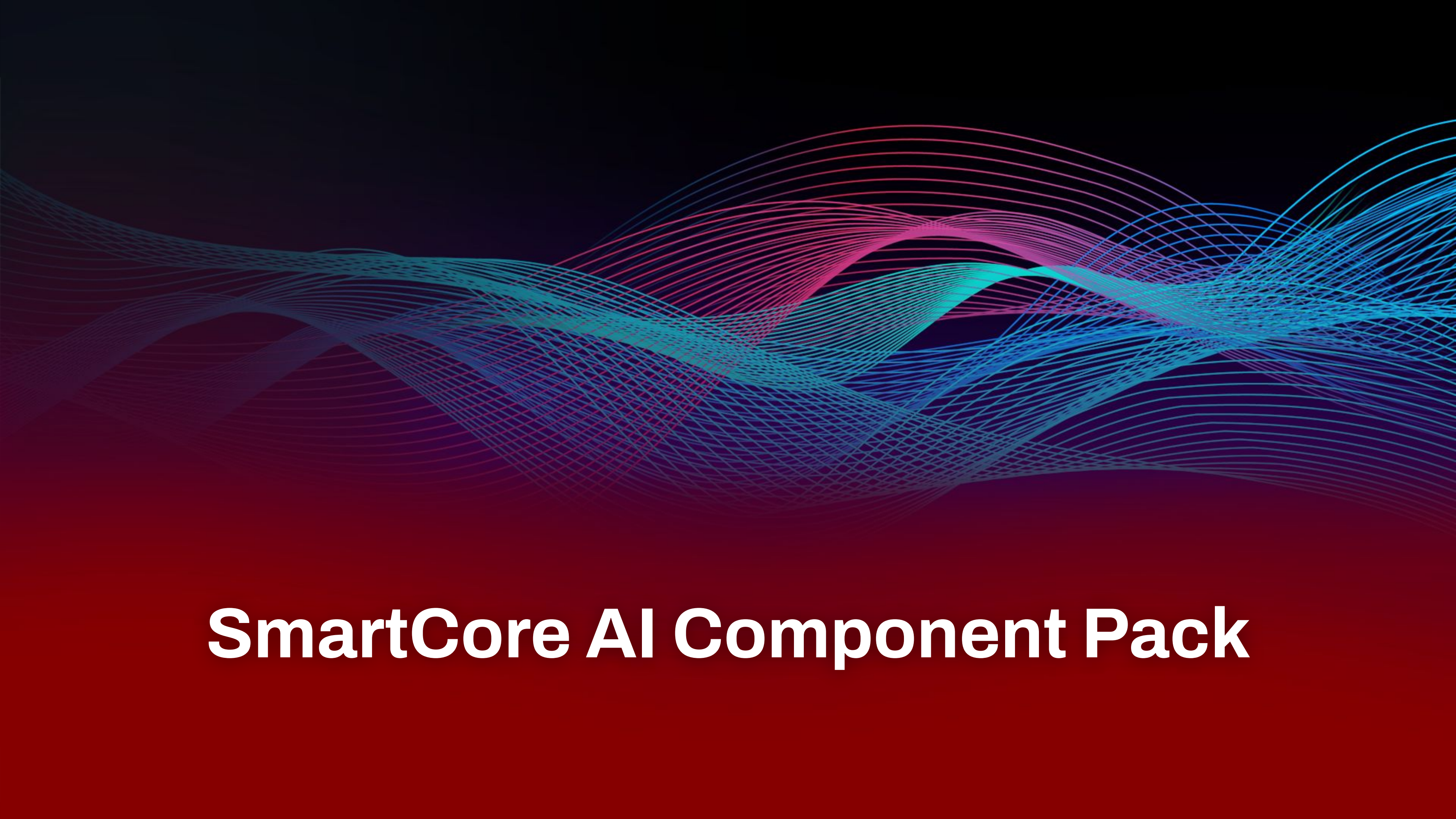
 Provide Feedback

Please enter your question...





RAD AI Companion



SmartCore AI Component Pack



SmartCore AI Components

Smart CodeInsight is our AI for use inside the RAD Studio IDE

SmartCore AI Component Pack is Your Gateway to AI inside Your Apps

A new set of core AI components designed to help our customers add AI features to their apps or build brand new AI-focused apps with Delphi or C++Builder – Available in GetIt

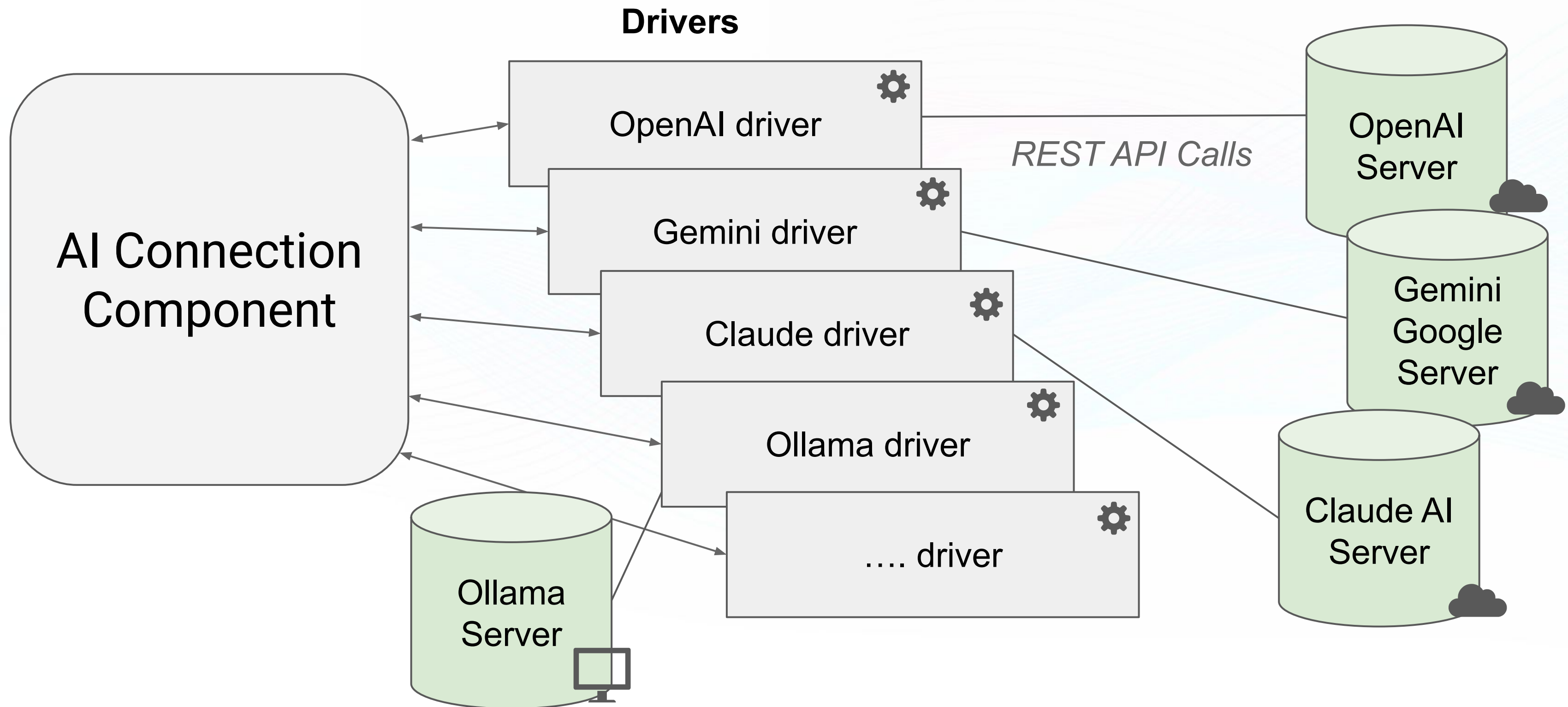


Smart AI Components Pack

- **A building block and a foundation for the future**
 - A core architecture our customers and 3rd party vendors can build on and extend to integrate with additional AI engines and UI controls
- **AI connection component (with a role similar to the database connection in FireDAC)**
 - Single AI-engine agnostic component uses specific drivers for AI engines
 - Includes drivers for OpenAI, Claude, Gemini and Ollama, you can add additional drivers for other AI providers
 - Configuration is managed via an easy-to-use AI connection wizard
- **A set of AI request components for different data types**
 - Like text, images, structure data
 - Can be associated with prompts and can be bound directly with UI controls to display the results
- **This component-based model makes it very simple to use the SmartCore AI Components and add them to existing VCL or FMX apps**



Smart AI Components Pack: Drivers Architecture



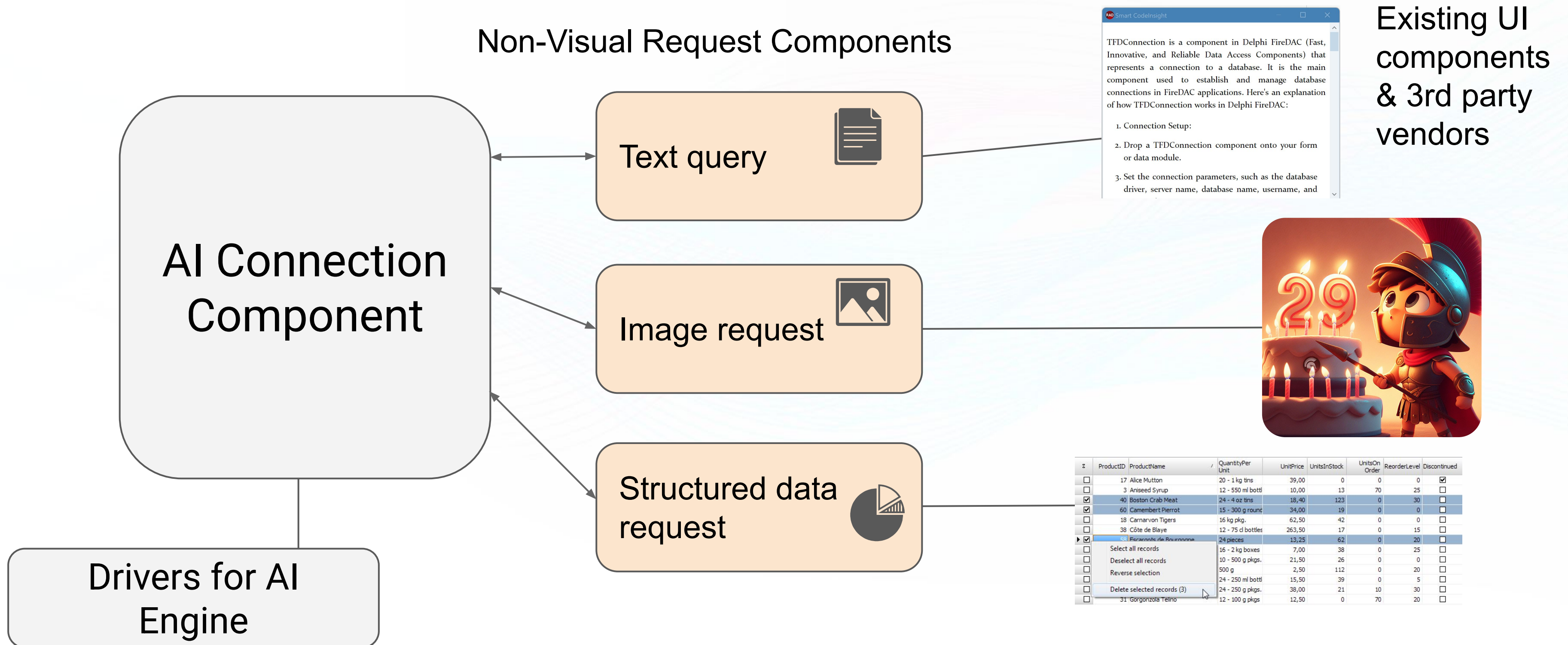


Smart AI Components Pack: AI Connection Component

- A single component regardless of the backend
- A set of internal drivers to map to different engines
- Expose the drivers' configuration to the connection, with key properties plus some variable ones
- Offer a “connection wizard” to configure a connection and test it out directly
- Possible to add new drivers for additional AI engines (similar to what we offer in the IDE via ToolsAPI for Smart CodeInsight)



Smart AI Components Pack: Frontend





Smart AI Components Pack: Request Components

- **“Request” components for different data types and processes**
 - Chat / text response supporting plain text and Markdown
 - Could also include translation requests
 - Image request
 - JSON / structured data request
 - Audio content or stream request
 - Upload audio and receive text
 - Upload text/query, receive audio
 - Upload audio, receive audio
- **Integrates with familiar UI controls: text, image, etc**



Smart AI Components Pack: Demos Preview

The screenshot displays the RAD Studio 13 Florence IDE interface. The main window shows a form with various AI components like `AIChatBindSource1`, `AIConnection1`, `AIOpenAIDriver1`, `BindingsList1`, and `AIChatRequest1`. The `LiveBindings Designer` pane shows a data flow diagram connecting these components. Two floating windows demonstrate the AI capabilities:

- VCL Image (OpenAI):** A window titled "VCL Image (OpenAI)" with a text input field containing "A cozy cabin by a lake at sunset, highly detailed, realistic." and a "Generate Image" button. Below the input is a generated image of a cabin by a lake at sunset.
- VCL JSON (OpenAI):** A window titled "VCL JSON (OpenAI)" with a text input field containing "Return array of any 5 cities with fields name (string), average_temperure (number), and number_of_citizens (number). Only JSON-array." and a "Send JSON Request" button. Below the input is a table of city data and a JSON response.

name	average_temperure	number_of_citizens
Tokyo	16	13929286
New York	12.5	8419600
Cairo	22	9500000
Sydney	18.5	5312163
Paris	11.5	2148327

```
{
  "id": "resp_68bfa00c49808196a85027cb3ed858ff04a8f1134110b5b8",
  "object": "response",
  "created_at": 1757388812,
  "status": "completed",
  "background": false,
  "error": null,
  "incomplete_details": null,
  "instructions": null,
  "max_output_tokens": null,
  "max_tool_calls": null,
  "model": "gpt-4o-mini-2024-07-18",
  "output": [
    {
      "id": "msg_68bfa00d21b88196a39c98e4e635ed8104a8f1134110b5b8",
      "type": "message",
      "status": "completed",
      "content": [
        {
          "type": "output_text",
          "annotations": [
            {
              "logprobs": [],
              "text": "\n\n \"cities\": [\n\n {\n\n   \"name\": \"Tokyo\", \n\n   \"average_temperure\": 16.0, \n\n   \"number_of_citizens\": 13929286\n\n }, \n\n {\n\n   \"name\": \"New York\", \n\n   \"average_temperure\": 12.5, \n\n   \"number_of_citizens\": 8419600\n\n }, \n\n {\n\n   \"name\": \"Cairo\", \n\n   \"average_temperure\": 22.0, \n\n   \"number_of_citizens\": 9500000\n\n }, \n\n {\n\n   \"name\": \"Sydney\", \n\n   \"average_temperure\": 18.5, \n\n   \"number_of_citizens\": 5312163\n\n }, \n\n {\n\n   \"name\": \"Paris\", \n\n   \"average_temperure\": 11.5, \n\n   \"number_of_citizens\": 2148327\n\n }\n\n ]\n\n }],
      "role": "assistant"
    }
  ],
  "parallel_tool_calls": true,
  "previous_response_id": null,
  "prompt_cache_key": null,
  "reasoning": {
    "effort": null,
    "summary": null,
    "safety_identifier": null,
    "service_tier": "default",
    "store": true,
    "temperature": 1.0,
    "text": {
      "format": {
        "type": "json_object",
        "verbosity": "medium",
        "tool_choice": "auto",
        "tools": [],
        "top_logprobs": 0,
        "top_p": 1.0,
        "truncation": "disabled",
        "usage":

```




Platforms, RTL, Data

Platforms

*RAD Studio targets **multiple platforms***

(primarily with Delphi as C++ Builder is currently for Windows)

*We maintain **low level APIs** and SDK tools integration with all of them*

RAD 13 sees a sheer number of platform improvements



Target Platform Recap

- **Windows (Delphi and C++Builder)**
 - Windows 11, Windows 10; Windows Server 2025 and 2022
 - Target both 32-bit and 64-bit, IDE required 64-bit Windows
- **Other platforms (Delphi only)**
 - iOS 18
 - macOS 14 Sonoma, macOS 15 Sequoia (Intel and ARM)
 - Android 16, 15, and 14 (Android API level 35)
 - Ubuntu Linux 24 and 22, RedHat Enterprise Linux (version 10), Windows WSL2



Platform Integration (Mobile)

- **Android target default remains API level 35, toolchain improvements**
 - Updated Java2OP (Java to Object Pascal) API headers translation tool, moving from Doclava to jparser
 - Updated several enhancements to the tools used for building and deploying apps on devices: manifestmerger, AAPT2, dexter, rclasser, bundletool, packager...
 - *Note:* Google Play's 16 KB page size requirement (enforced later this year) was already supported in RAD Studio 12.3
- **iOS API cleanup and enhancements**
 - Added a couple of dozen missing functions in the core and different kits



Platform Integration (Windows)

- **Windows APIs cleanup, including:**
 - Updated WinRT APIs
 - Refreshed the classic WinAPI WinMD projection
 - Updated the WebView 2 control integrated in TEdgeBrowser
- **All of these updates provide an enhanced experience for Windows application developers**

Delphi RTL

***Delphi Run Time Library** (including the database RTL) serves as a companion of the compiler and a foundation for all other libraries
C++ has its own RTL, but it also leverages the Delphi RTL*

*RAD 13 focuses on RTL **quality**, offers many small enhancements*



Delphi RTL

- **JSON processing**

- New attribute to ignore a field in `TJsonValueSerialization`
- JSON Serializer offers the option to skip properties
- Improved `TJSONValue.GetValue<T>`
- `TJSONNumber` class reworked (but compatible)
- `TJsonReader.ReadAsUInt64`

- **Dozens of small RTL improvements**

- `TPath.Exists`, `TRegistry.ReadUInt64`, `TThreadedQueue.Count`, `GetStorageSize` for RTTI, `TSizeF.Zero`, `GUIDToBuf` function and more `GUIDToString` changes, `TMultiReadExclusiveWriteSync.WriterID` property... and many others

- **Web Client libraries**

- Added HTTP/3 protocol support to `THttpClient` via `THTTPProtocolVersion.HTTP_3_0`
- For SOAP clients, new asynchronous version of `THTTPReqResp.Execute`



Delphi Database RTL

- **Expanded support for Unsigned Big Integer database fields**
 - `TFieldType.ftLargeUInt` type
 - For Oracle, MySQL, as well as the `TFDMemTable`
- **Dragging fields to form**
 - More flexible options when dragging from the DB field editor and generating custom dataware controls by third-party vendors
 - New overloaded `DrpCtrls.RegisterFieldControl` procedure
- **And more**
 - New `EllipsisPosition` property added to the VCL `TDBText`
 - New `DataSet` properties to access the attached `DataSource` objects



FireDAC

*FireDAC is RAD Studio's primary **universal database access** library, offering support for all the most popular RDBMS*

In RAD 13, FireDAC continues to add support for new features in various databases and their drivers



FireDAC drivers

- **Additional updates to FireDAC drivers**

- OAuth 2.0 authentication for Oracle 19.13+
 - Use the `T0Auth2Authenticator` component
- Parallel backup operations in `TFDFBBackup` for Firebird 5.0+
- Improved support for PostgreSQL array types

- **Core FireDAC**

- `TFDMemTable` supports `RecsSkip` and `RecsMax` when filtered (for easier paging)
- `TFDMemTable` allows loading standard JSON
- Added a number of virtual methods to `TFDCustomConnection` and `TFDCustomTransaction` to help with custom subcomponents development



InterBase 15 is Coming

- RAD Studio bundles InterBase Developer and the InterBase ToGo embedded database for all target platforms
- RAD 13 still bundles InterBase 2020, but it already includes a license for the upcoming InterBase 15 Developer edition





UI Libraries



VCL

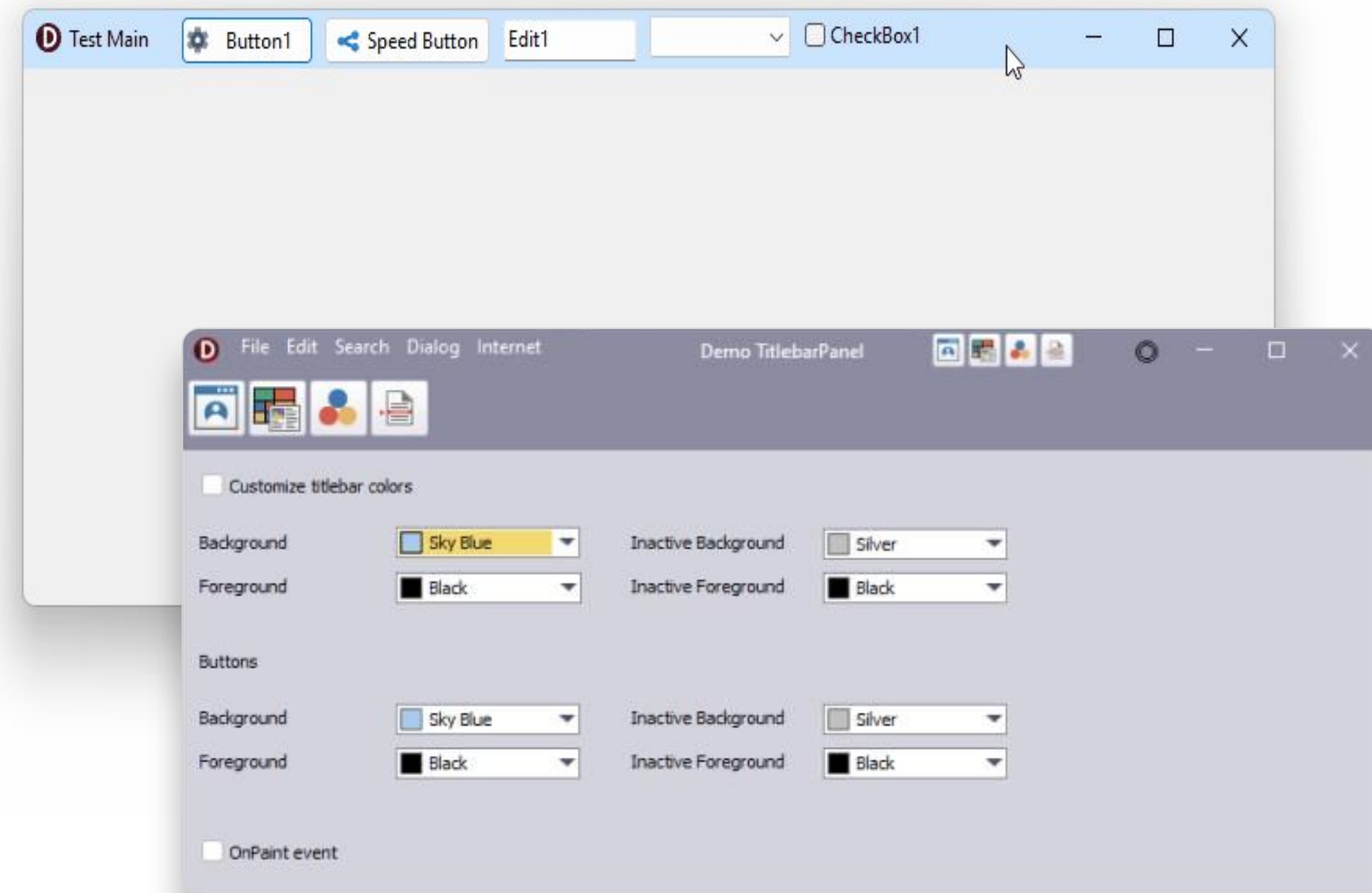
*The VCL library is the **best** and most comprehensive **UI components** library **in the industry** for building modern Windows desktop applications. Embarcadero continues to invest in VCL, which remains at the forefront of RAD Studio*

RAD 13 adds a couple of long awaited feature extensions to VCL



VCL

- Support for VCL styles for this TitleBar (TitleBarPanel component), a feature requested since the TitleBar was introduced in the VCL library
 - Plus new `TTitleBar.StyleColors` property to enable automatic use of the VCL Styles colors for the background and the buttons on the title bar
 - Controls you can use on the TitleBar: `TButton`, `TSpeedButton`, `TCheckBox`, `TRadioButton`, `TToolBar`, `TEdit`, `TComboBox`, `TFormTabsBar`, `TActionToolBar`, `TActionMainMenuBar`
 - New `TCaptionButtonItem` `ShowHint` property

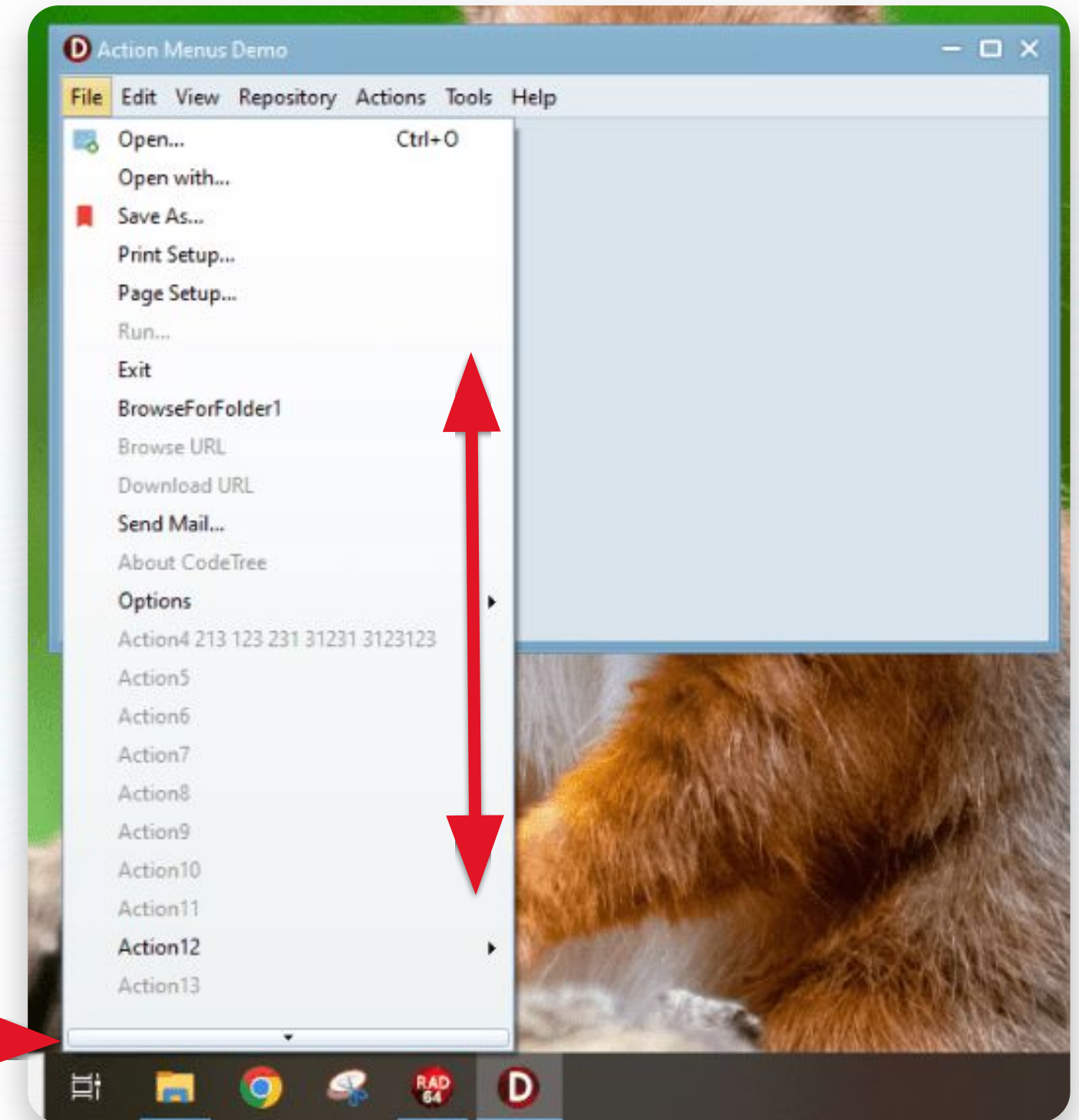




VCL

- New scrolling feature for the ActionMainMenuBar control
 - Enabled when the vertical height exceeds the screen height
 - Works for submenus
 - Works for PopupActionBar local menu
 - Plus support for menu items hints in the ActionMainMenuBar

See also <https://youtu.be/wLq1htj8mrM>



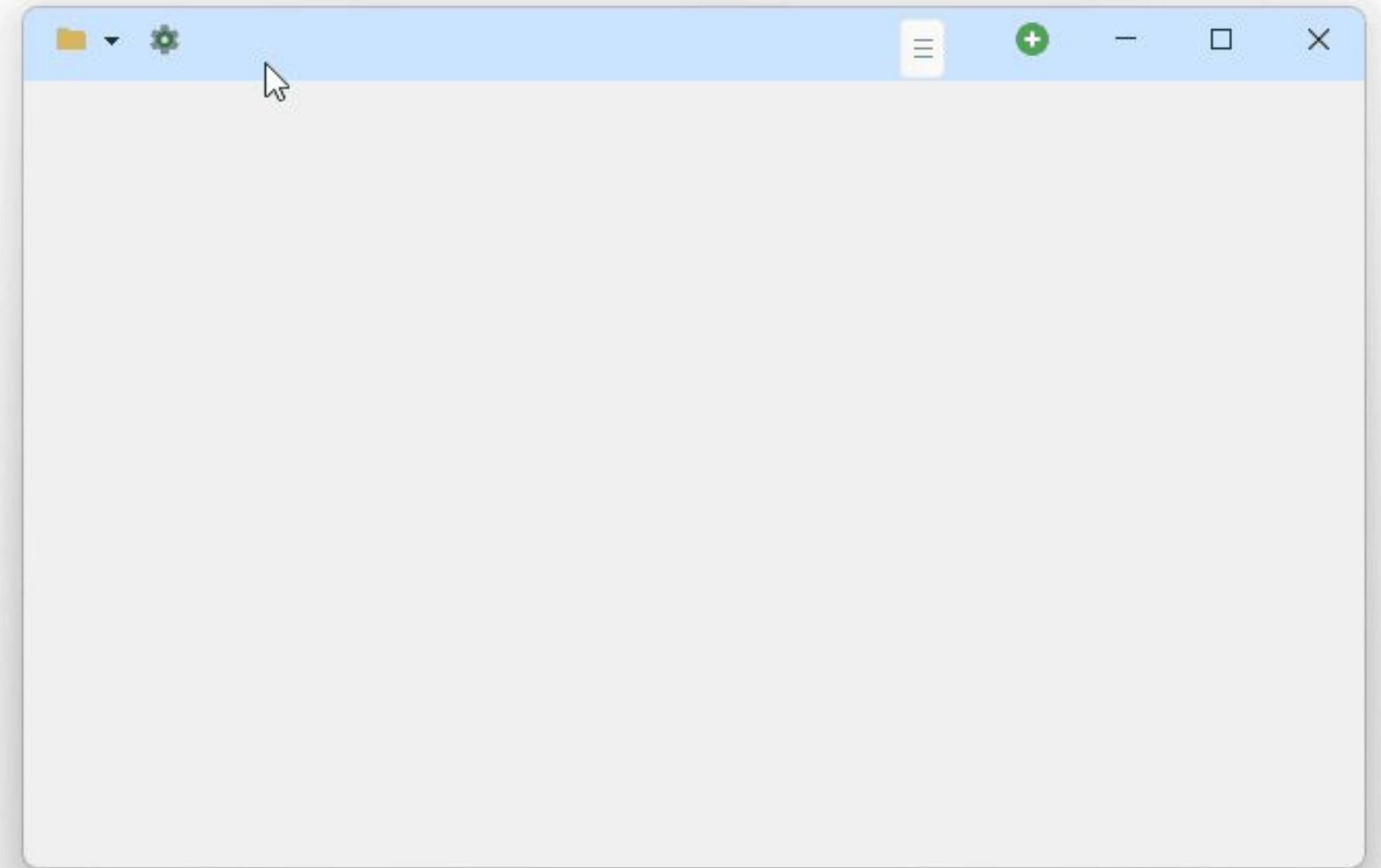


VCL

- **FormsTabsBar improvements**

- Transparency (Transparent and ShowBottomLine)
- OnGetTabCaption event, MouseMiddleButtonClose property
- Improved MDI support

Makes MDI application look modern!





VCL

- Integration of the UIAutomation Microsoft interfaces, which can be used to support accessibility and to build automatic UI testing
 - Core APIs, RTL support, `TWinControl` integration, specific implementation classes
- Additional improvements
 - Further work on the very powerful custom `ControlList` component
 - Modernized `TToggleSwitch` UI
 - Improved `TEdgeBrowser` component (WebView 2 control)
 - WinUI 3 Demo part of the core product demos
 - New `TStyleManager.FormBorderSize`
 - New `TMonthCalendar.OnChange` event
 - New `OnBeforeResize` and `OnAfterResize` events for `TSplitter`
 - And much more



FireMonkey

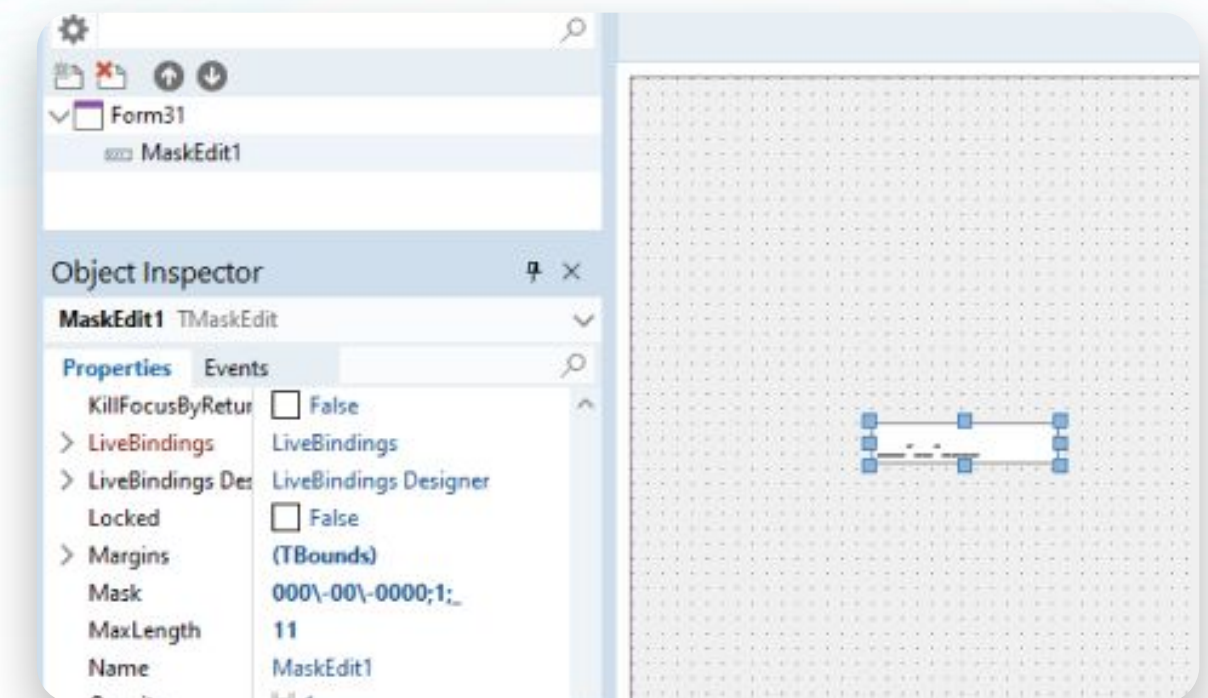
*FireMonkey is the RAD Studio **multi-device** library, allowing you to use Delphi to build **mobile** and **desktop** applications for 5 operating systems from a **single source code** base*

*In RAD 13, FireMonkey has a new architecture for **fast animations** and **new UI controls** and components*



FireMonkey

- **Display Link Service** - Smoother animations using platform-native APIs: JChoreographer (Android), CADisplayLink (iOS)
- **New FMX MaskEdit component** - VCL-style input masking
- **TPresentedScrollBar enhancements** - Control scroll animation, bounces, and touch interaction
- **Enhanced spell checking** - New IFMXSpellCheckerServiceEx interface with redesigned service
- **TAlignLayout improvements** - Added center alignment options
- **TApplicationEvents component** - New event handling capability
- **TComboEdit AutoComplete** - Built-in text completion support
- **GPU-accelerated bitmap copying** - Significantly faster performance

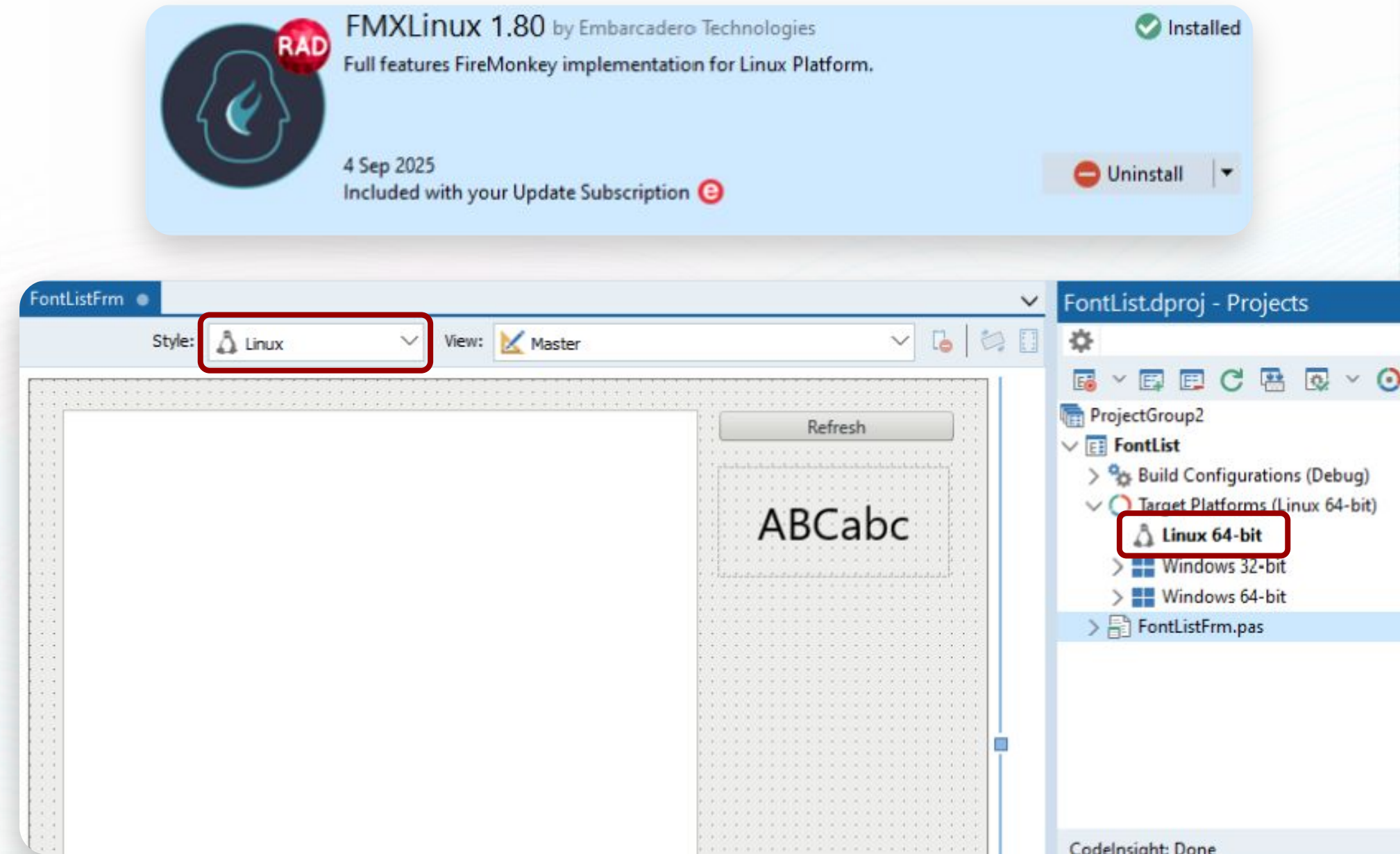


MaskEdit control



FireMonkey

- Improved **FMXLinux** integration
 - Making it easier to use the Linux UI (including FMX view at designtime)
 - Available for Enterprise customers as a GetIt package

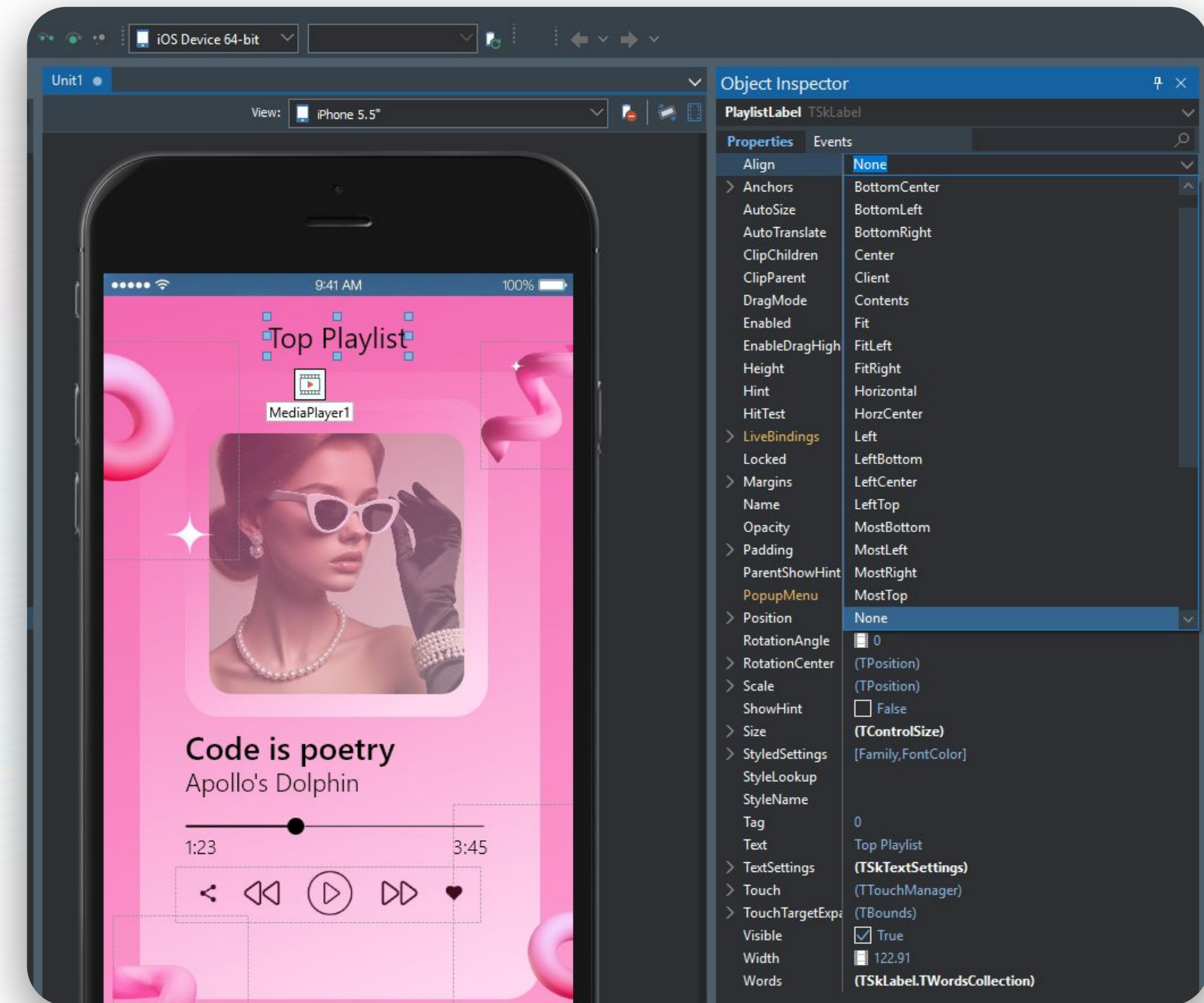


- **Skia4Delphi** support updated to the most recent version of the library
- Updated WebBrowser component for Windows



FireMonkey

- Support for the 16.0 version of Emoji symbols
- New OnCheckedChange event for TListBoxItem
- New TTrackBar Reverse property (and mouse wheel scrolling support)
- MenuItem height can be styled
- Mobile
 - Fullscreen support for iOS
 - Edge to edge support for Android
 - Ability to disable the bounce in a list view on iOS





WebBroker, WebStencils, RAD Server



WebBroker

WebBroker is RAD Studio's framework for building web server extensions for IIS, Apache, or Nginx (new in 13) and stand-alone servers. WebBroker is the foundation for RAD Server, DataSnap and third party libraries

*RAD 13 introduces the **largest upgrade** to WebBroker since **Delphi 6!***



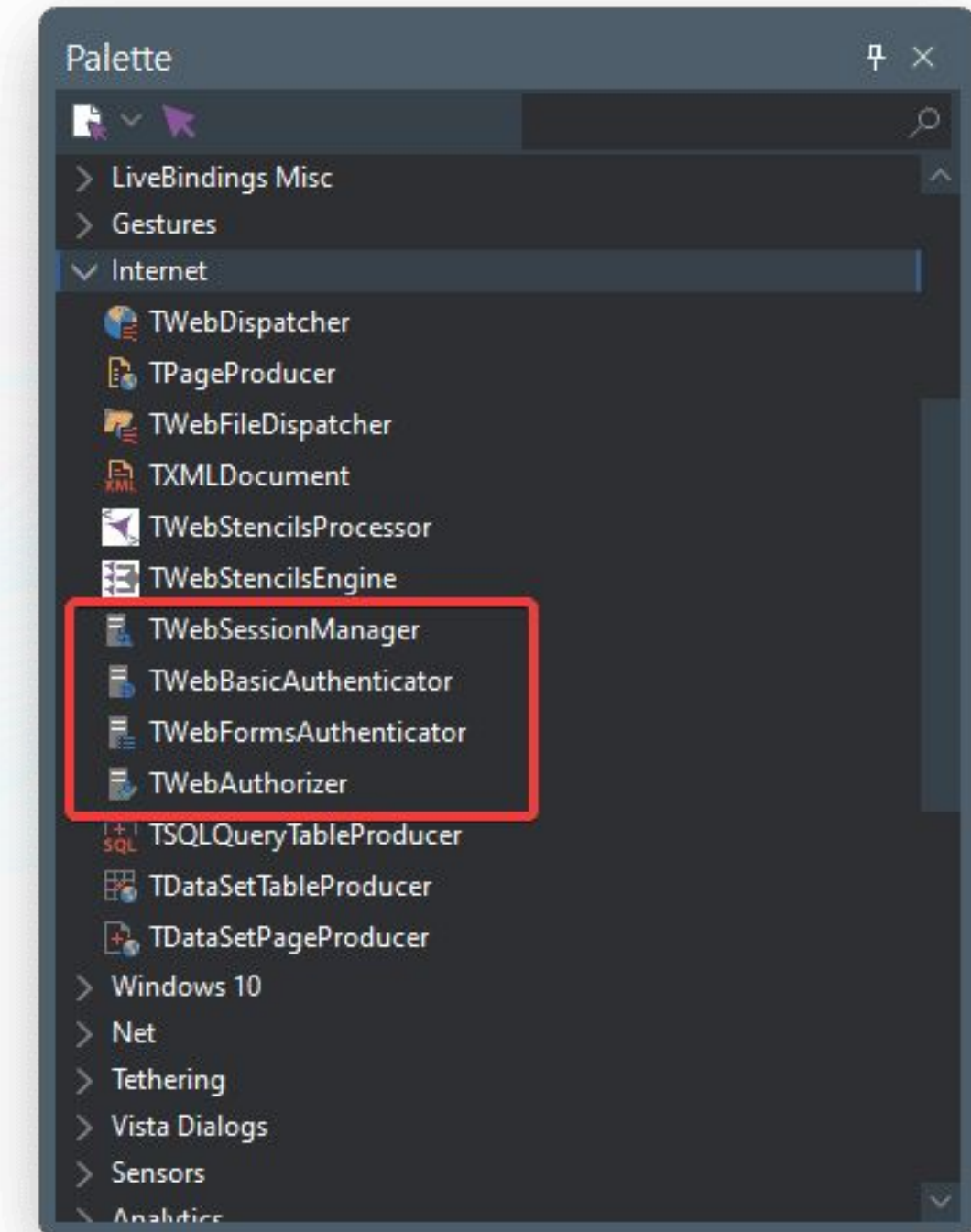
WebBroker Enhancements (General)

- **Integrated Sessions**
- **FastCGI support**
 - For Nginx web server (and Apache)
 - New classes `TFastCGIApplication`, `TFastCGIRequest`, `TFastCGIResponse`
 - Support for FastCGI added to the WebBroker Wizard
- **Support for Apache Config**
 - A textual, hierarchical, and extensible configuration system
 - Uses mapping of a Delphi record to Apache directives
- **Integrated logging**
 - `TWebApplication.Log` method (with severity, message, params)
 - Mapped for most configurations
 - Apache server log file, `logs\error.log`
 - ISAPI: Windows Event Log (`eventvwr.msc`)
 - CGI: file with `TCGIApplication.LogFileName` name
 - FastCGI: HTTP (Apache, nginx) server log file, `logs\error.log`



WebBroker Sessions Management

- **TWebSessionManager** class with list of **TWebSession** objects
- A SessionId cookie by default (other options available)
- New **Session**, **AuthMethod** and **AuthUsername** properties in **TWebRequest** class
- Ready-to-use authenticators
 - **TWebBasicAuthenticator** ("Basic" HTTP authentication)
 - **TWebFormsAuthenticator** (HTML form-based authentication)
- **TWebSessionManager** non visual component for configuration
- **TWebSession** object includes DataVars (session data), User, base information like CreatedTime, LastURL, LastIP
- **TWebAuthorizer** provides role-based access control through authorization zones





WebStencils

*WebStencils (a recent new feature of RAD Studio) extends the existing web technologies in RAD Studio (WebBroker, DataSnap, RAD Server) by providing a **server-side template engine**, for building efficient **data-driven websites***

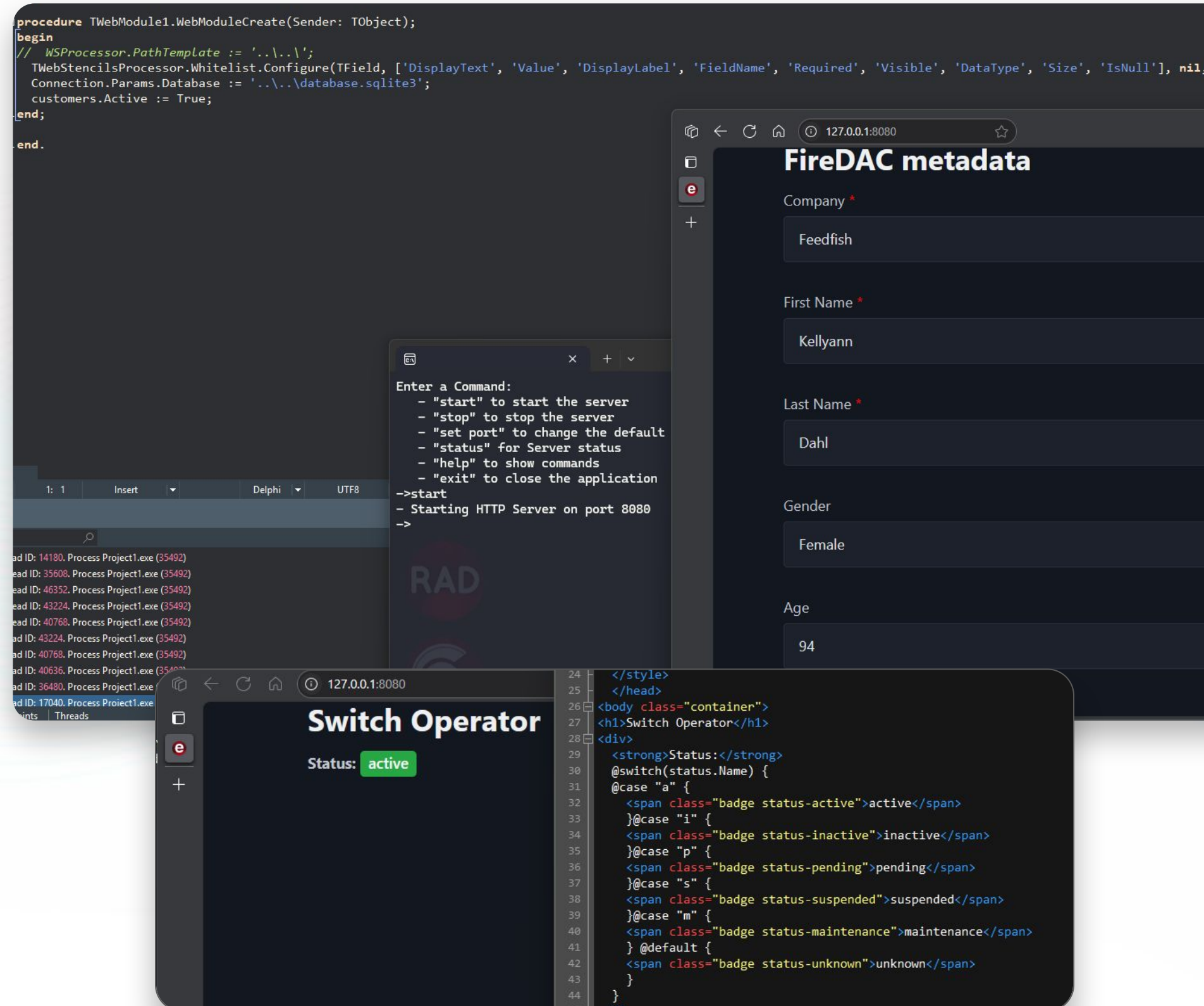
*In 13 WebStencils gets a **major upgrade** with session, new data access, a switch statement, and much more*



WebStencils Enhancements

New WebStencils features in 13 Florence

- Support for session data
- Extra Variables outside of the class/data model (via an anonymous method)
- New **@switch** statement with cases for different string values and expressions or **@default**
- Direct access to dataset properties
 - Along with new allow/disallow security lists to precisely control what scripts can or cannot access
- Nested **@ExtraHeader** Support
- Extra properties in **@page** keyword

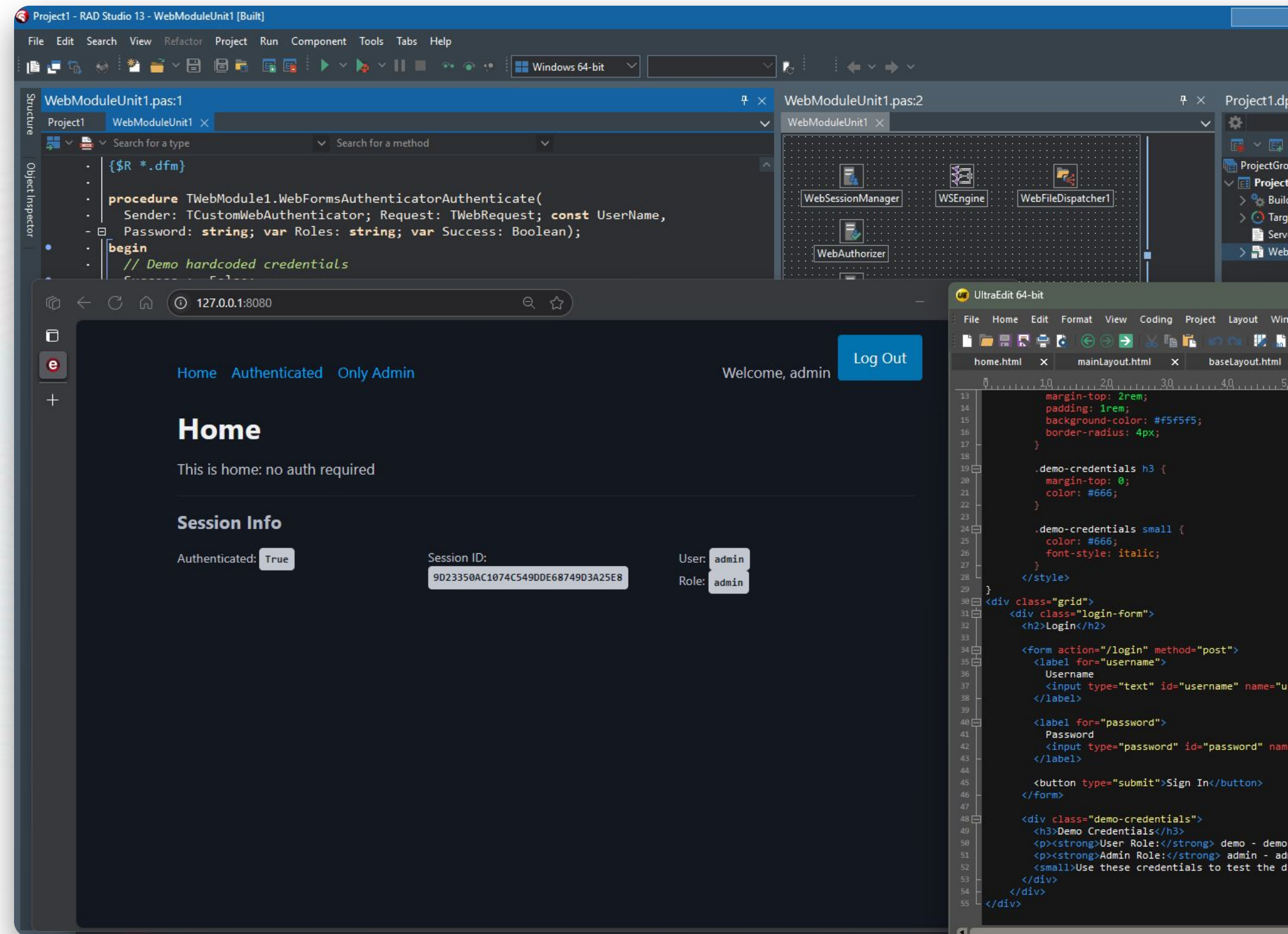




WebStencils and Sessions

Sessions integrate with WebStencils templating:

- **@session.xxx**
Access session object properties
- **@session.DataVars**
Access session custom variables
- **@session.UserHasRole(<string>)**
Check if a user has a specific role
- **@<name>**
Direct access to objects stored in DataVars with key <name>

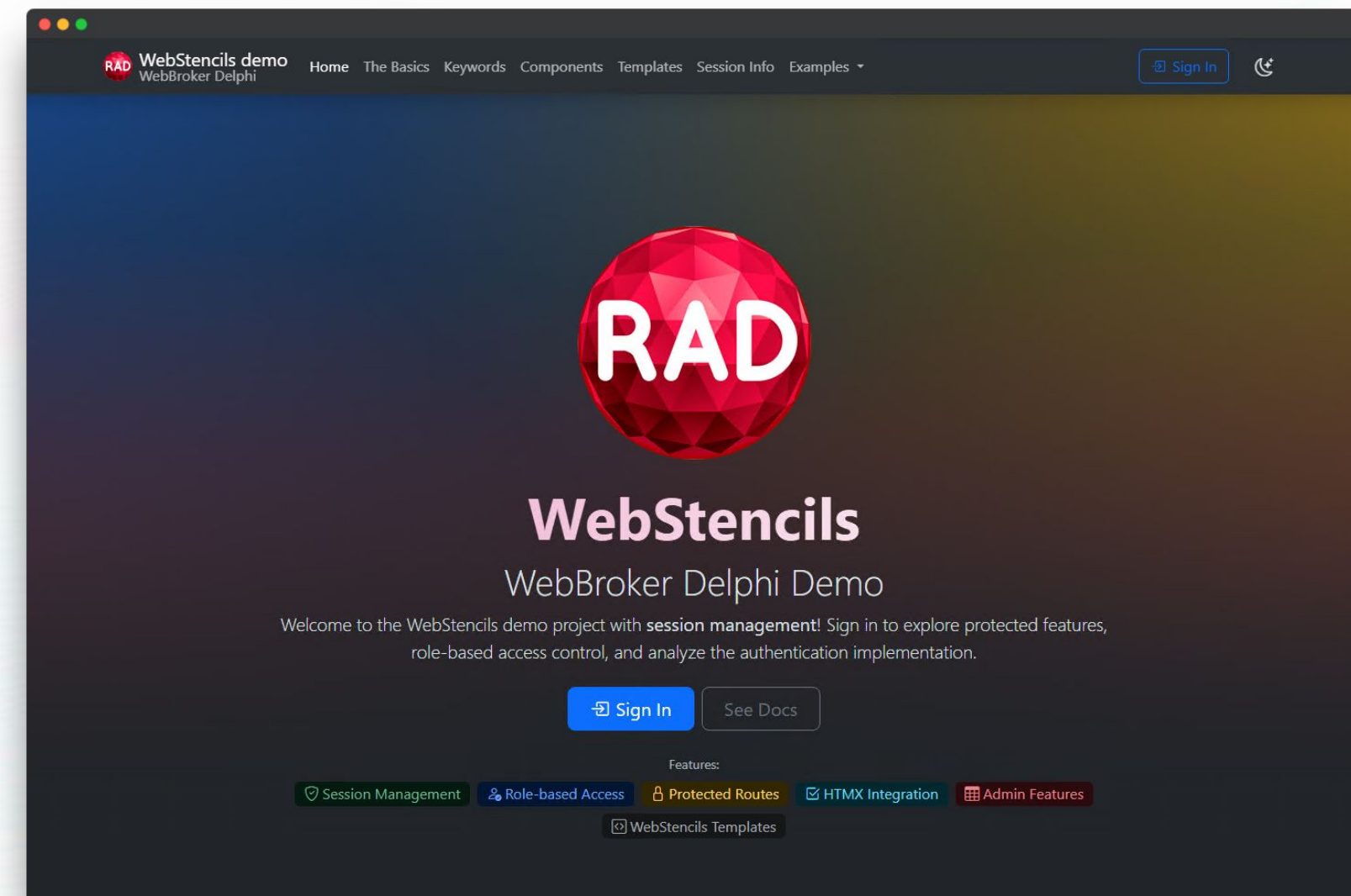




WebStencils is More Powerful in 13.0

- WebStencils is **even easier** and more **powerful** than in 12.2
 - Now includes a solid foundation for adding common features like sessions, user logins, and permissions management
 - Easier to deploy and better integrated with more web servers (Nginx)

*Building **powerful**, fast and **robust web** servers powered by your business data is getting much **faster** with RAD Studio.*



RAD Server

*RAD Server is Embarcadero's **modern, middle-tier** solution for publishing **REST APIs** for client applications. It's a **ready-to-use** engine you extend with packages with specific URL mappings*

*RAD Server is getting continuous updates and improvements
(and it's available in Enterprise+ SKUs only)*



RAD Server Enhancements

- More flexible Swagger-based API Documentation
 - Schema and reference parameters are now optional
 - New SetBaseUrl and GetBaseUrl methods in IEMSApiDocumentService
- RAD Server Multi-tenancy adds an option to integrate tenant-specific configuration
 - Tenant-prefixed server variables
 - BeforeRequest and AfterRequest in IEMSResourceInterceptor interface
- Introduces a simple and direct way to implement API Versioning
 - That is having a server offering multiple versions of the same REST APIs for older and newer client applications
 - With a new ResourceVersion attribute

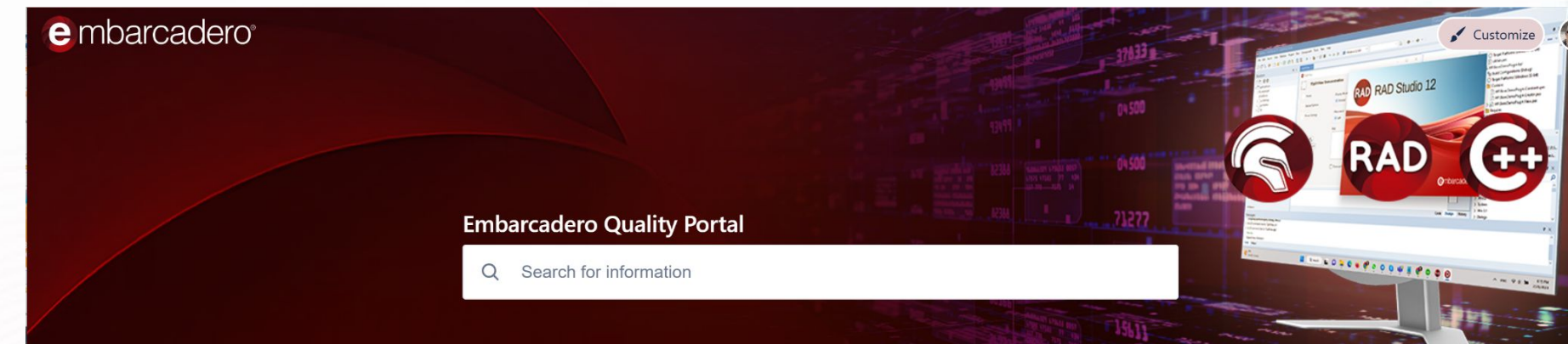


But What About Quality?



Public Customer Reports in 13.0

- 318 customer reported issues fixes
- 108 customer feature requests implemented
- Overall, **734** customer reports addressed with various resolutions in the two customer portals combined
 - Old portal (quality.embarcadero.com)
 - New Quality Portal (qp.embarcadero.com)





RAD 13 Florence Summary

Delphi: Ternary operator and More

Clang 20 upgrade for C++ Builder

Complete 64-bit IDE for Windows 64-bit platform

VA & code tooling improvements

SmartCore AI Component Pack

VCL TitleBar Styling and new FMX controls

WebBroker and WebStencils extensions

RAD Studio 13 is Available Today





< RAD 13 Florence >

Now Available!

Download



Questions?